# A TEARDOWN OF APPLE'S ON-DEVICE FOUNDATION MODELS: ARCHITECTURE, QUANTIZATION, AND CONTAINER FORMATS OF THE `FM_GENERATIVEMODELS` ASSET SET

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Abstract. We present a static teardown of Apple's on-device generative-model asset bundle, `com.apple.MobileAsset.UAF.FM.GenerativeModels`, as shipped to a macOS 26/27 machine, together with an attempt to reconstruct a running forward pass from the recovered tensors. The catalogue holds 124 signed assets across six model families; we document the container stack (Cryptex images, the `BBBB` weight container, the `odix` executable, the ANE `.hwx` binary, the MPSGraph package), the `odix` reference scheme, and two language backbones—a dense ∼3 B model and a sparse Mixture-of-Experts with instruction-following pruning—plus ∼90 task adapters, guardrail models, draft models and server variants.

The weight codec is 4-bit LUT palettization with a per-block fp16 scale under an ANE tile permutation. We reverse it, close the parameter budget exactly against the physical file (9.862 B parameters, 100.00% of the shipped weight bytes), and validate the decoder by round-trip against known weights at correlation 0.981—the 4-bit quantization floor. Two obstacles previously believed fatal dissolve: the router→expert map is mathematically irrelevant to the output because the pruned feed-forward is an ungated permutation-invariant sum, and the "missing constant table" has a shipped runtime equivalent. A later OS build additionally ships the 3B token embedding that earlier builds withheld.

A coherent from-weights forward pass nevertheless does not follow. We localize the obstruction to a single ANE storage convention—the coefficient ordering used by residual-writing projections—which no available compiler reproduces for an isolated operator. We also retract four of our own intermediate results, each traceable to a null hypothesis that destroyed more structure than the hypothesis under test; we consider that failure mode the most transferable finding in this work. The analysis is read-only and no signing material was bypassed; no weights or tokenizer data are reproduced here or published.

## Executive Summary

This report documents a static teardown of the on-device model assets that ship with Apple Intelligence, together with an attempt to reconstruct a running forward pass from the recovered tensors. It is written to be read in either direction: as a description of the shipped artefacts (§13–§7), or as a record of which reconstruction claims survived scrutiny and which did not (§8).

**What the assets contain.** The `UAF.FM.GenerativeModels` catalogue holds 124 signed assets spanning six model families. The weight codec is 4-bit LUT palettization with a per-block fp16 scale, stored under an Apple Neural Engine tile permutation. Both are reversible: the parameter budget closes *exactly* against the physical file (9.862 B parameters, 100.00% of the shipped weight bytes, zero non-finite tensors), and the decoder is validated by round-trip against known weights at correlation 0.981—the 4-bit quantization floor.

**What that means in practice.** The shipped assets are sufficient to recover the model weights, the architecture, the tokenizer, the router, and—as of macOS 27.0 build `26A5378n`—the 3B token embedding, which earlier builds did not ship. Two obstacles that were previously believed fatal turned out not to be: the instruction-following-pruning router→expert map is *mathematically*

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*irrelevant* to the output (Finding 9), and the "missing constant table" has a shipped runtime equivalent (Finding 10).

**What still does not work.** A coherent from-weights forward pass. The obstacle is the element→position *ordering* of the recovered tensors, and §10 reports it as unresolved for every weight role rather than for one. The file format itself is settled: the tile payload begins at byte +96, not +128—a real decode defect found by its cause and proven structurally, since all examined tiles carry exactly zero in their final 32 bytes, and confirmed on a second independent build whose weight map reproduces the first *bit-exactly* (Finding 32). The seven-role decomposition and every tensor shape are confirmed from Apple's own shipped `manifest.plist`. All recovered tensor *values* are trained matrices. Only their arrangement is unknown (Observation 3).

**On negative results.** This is the most transferable part of the report. Instruments used during this work repeatedly produced confident conclusions that later failed—first weight statistics whose null hypothesis destroyed more structure than the hypothesis under test (Findings 20, 23, §8.4), then a composition $z$-score (§9.7), then a likelihood-based replacement. The last of these is characterised rather than merely withdrawn: applying the best surviving candidate ordering to one layer at a time improves 14 of 24 layers, and a *random* permutation improves 14 of 24 (Finding 33). Perturbing the axis at random helps half the time, which is the signature of a tensor already contributing noise—so no ranking among permutations carries information, and the earlier claim that six of seven roles were positively confirmed does not survive. Ground-truth activations would settle the question directly, and §10 reports that route closed on measurement as well: the values never enter user-process memory, and the runtime logs that would expose the layout are redacted behind a signing gate.

**Findings at a glance.**

| # | Finding | Status |
|---|---|---|
| 1 | Weight codec is 4-bit LUT palettization × per-block fp16 scale | Established |
| 2 | Parameter budget closes exactly against the physical file | Established |
| 3 | Codec is necessary, then sufficient, for reconstruction | Established |
| 4 | Budget closure bounds the search fan-out | Established |
| 5 | The ANE spatial permutation, reversed | Established |
| 6 | The `.hwx` program holds no weights | Established (negative) |
| 39 | `odix` reference scheme: self-relative signed offsets | Established |
| 7 | Recovered weights cross-validated against Apple's own toolchain | Established |
| 8 | Per-layer norm $\gamma$ is compile-time folded | Established |
| 9 | Expert-selection map is a *false* blocker | Established |
| 10 | "Missing constant table" = the shipped `metadata.bin` | Established |
| 18 | Deployed sparse-FFN shape recoverable from `metadata.bin` | Established |
| 11 | Intermediate activations are ANE-internal | Established (negative) |
| 14 | Dynamic capture localizes the live embedding | Established |
| 15 | Host capture recovers the small model's embedding | Established |
| 21 | Real input is lowercased, chat-templated, guard-model-preceded | Established |
| 24 | ANE coefficient z-order, positionally read, round-trip validated | Established |

| # | Finding | Status |
|---|---|---|
| 25 | OutTrans=1 marks the residual-writing projections | Established; remainder open |
| 12 | Embedding absent from the weight file | Superseded (17) |
| 13 | 3B ships no embedding file | Superseded (17) |
| 16 | 3B embedding is ANE-locked | Superseded (17) |
| 17 | A later OS build *ships* the 3B embedding | Established; layout open |
| 19 | Full-forward ablation: KV geometry, ungated sum | Established; oracle caveated |
| 20 | Single-token rank oracles manufacture false positives | **Retraction** |
| 23 | A weight-statistics test decisive on Qwen is vacuous on AFM | **Retraction** |
| 22 | The pico forward's defect is one block | Amended (23) |

**Open problems.**

| Problem | Why it is open | Kind |
|---|---|---|
| OutTrans=1 coefficient order | Chosen by the ANE scheduler from full-graph context; no available compiler emits it for an isolated operator | Tooling limit |
| 3B embedding element layout | File present and exactly sized; row stride and 8-lane structure proven, per-token vectors not yet recovered | Decode problem |
| 56,064-wide FFN alignment (3B) | No per-layer ground truth: activations never leave the ANE | Information limit |
| Which-46-of-219 pruning list | Consumed at export time; not in any shipped asset. Irrelevant to output correctness (9) | Information limit |
| True layer count of the draft model | 38 coefficient blocks beyond the base architecture form a KV-reusing layer-shaped unit after the last mapped layer | Open |

**Method and scope.** All figures come from mounting the shipped, unencrypted Cryptex disk images read-only and parsing their contents. No signing material was bypassed, no protection was circumvented, and no code was executed on the models to obtain the static results. One step is dynamic: a user-privileged `lldb` core dump of the reader's own running process on their own hardware, used to observe an embedding buffer that the CPU produces in host memory (§15).

No model weights, tokenizer data, or other Apple-proprietary artefacts are reproduced in this document or in the accompanying public repository, which contains only original analysis and decoder source. Quantities reported here are measurements, offsets, and statistics—not content.

## 1. Introduction

Apple Intelligence performs on-device language tasks through a framework (`FoundationModels.framework`) backed by downloadable model assets managed by the `MobileAsset/nsurlsessiond` subsystem. This report inventories and dissects one complete copy of the `UAF.FM.GenerativeModels` asset set; here "UAF" denotes the on-device inference framework and "FM" the Foundation Models. Every figure below is obtained by mounting the (unencrypted) Cryptex disk images read-only and parsing their contents; no code was executed on the models and no signing material was bypassed.

## 2. Asset Distribution System

Each asset is a directory `<sha1>.asset/` under the bundle `com.apple.MobileAsset.UAF.FM.GenerativeModels`, containing:

- `Info.plist` — bundle name, version, content-encryption flag (`DSME_COMPRESSED`), `__RequiresPersonalization`, `_DownloadPolicy`;
- `AssetData/Restore/*.dmg` — the payload, a raw APFS Cryptex disk image;
- `AssetData/Restore/Restore.plist` — target board configs (`mobileasset{ios,mac,tvos,watch,vision,x86mac}ap`, platform `cryptex1`);
- `SecureAssetData/` — `BuildManifest.plist`, `*.root_hash`, `*.trustcache`, `*.integritycatalog`, an Image4 ticket (`apticket.*.im4m`) and TSS request/response.

Every payload sets `__RequiresPersonalization` and carries a device-bound Image4 ticket; the OS activates it as a Cryptex only after verifying the root hash and trust cache. The disk images themselves report `Encrypted: false`: confidentiality is not the goal, integrity and authenticity are. The inspected cache holds 118 assets totalling ≈10.0 GB of disk images; versions span `0.0.2` to `14.5.x`, with the bulk at `14.x` (build `13M204130/13M204216`).

## 3. Model Families

Table 2 groups the assets; six families are present. The catalogue held 118 assets when first inventoried and 124 on macOS 27.0 build `26A5378n`; §17 covers the delta.

| Family | # | Role |
|---|---|---|
| `fm.language.instruct_3b` | 60 | Main LM (nano dense + ifp MoE) plus adapters |
| `gm.server.instruct_server_v1` | 28 | PCC (server) task adapters |
| `fm.language.instruct_300m` | 21 | Safety, classifiers, voice control |
| `gm.translate_fm` | 2 | Machine translation (alignment + phrasebook) |
| `gm.server.instruct_server_small` | 2 | Server safety / abuse guardrail |
| `fm.language.instruct_9m` | 2 | Text-event-extraction classifier |

**Table 2.** Model families in the asset set.

## 4. The instruct_3b Backbones

Two *different* base networks share the `instruct_3b` name.

4.1. **Dense `afmplus-v11.0-nano`.** From `metadata.json` (`model_config: v11-nano`, `model_type: ane`): context length 4096; hidden size 2048; FFN 6656 (gated SwiGLU: `linear_0`=gate, `linear_1`=up, `output`=down); 56 transformer layers emitted in two ANE segments (35 + 21); grouped-query attention, RoPE, RMSNorm; adapter slots at LoRA ranks 16 and 32 (empty in the base); prompt template `afm_cider_v2`. The `base.generic` image is 1.3 GB, i.e. ≈2 bits/weight.

4.2. **Sparse MoE `afmplus-v11.0-ifp`.** The `base.generic_sparse` (`IFP`) image (5.8 GB) is a Mixture-of-Experts network. Its `ifp/config.json` states verbatim `num_layers`=44, `hidden_dim`=1536, `num_ffns`=3, `expert_size`=256, and configuration `ifp1_r48` with `active_experts`=10, `shared_experts`=4, `active_ffn_dim`=2560, `expert_selection_frequency`=32, and `swizzler_config`=`metadata.bin`. Table 3 gives the recovered structure and Figure 1 the per-layer data flow.

*Observation* 1. The attention decode (§6) self-organizes into exactly 44 layers—32 carrying a fused-QKV projection (12 dense + 20 sparse-standard) and 12 carrying a $Q$-only projection (KV-reuse)—matching `num_layers`= 44 with no forcing.

| Property | Value |
|---|---|
| Model dim $d$ | 1536 |
| Layers | 44 = 12 dense + 32 sparse |
| attention split | 32 fused-QKV + 12 KV-reuse (Q-only) |
| Attention | $Q = 2048$ (16×128), $KV = 1024$ |
| GQA ratio | 16 query : 4 KV heads, dim 128 |
| QK-norm | per-head RMSNorm on $Q$ and $K$ |
| Residual norms | pre *and* post, attention and FFN |
| Positional | RoPE (`cos/sin_cached`) |
| Dense FFN | SwiGLU, intermediate 3072 |
| MoE FFN | 10 active + 4 shared, freq. 32 |
| Embedding | int scalar-quant (scale + zero) |
| Output | tied embedding + `output_norm` |
| Baked adapters | LoRA rank 48 (`lora_i1_r48`) |
| Context | 8192 |
| Vocabulary | 262,144 (§11) |
| Origin framework | `tamm` / `coreflow` |

**Table 3.** Recovered `afmplus-v11.0-ifp` MoE architecture.

| Model | Layers | Total params | Active/token |
|---|---|---|---|
| `afmplus-v11.0-nano` (dense) | 56 | ≈3.0 B | ≈3.0 B |
| `afmplus-v11.0-ifp` (MoE) | 44 | 9.862 B | ≈1.4 B |

**Table 4.** Parameter budgets. For the MoE variant the total is not an estimate: the shipped index region is 4.931 GB of 4-bit weights = 9.862 B parameters exactly, of which our export recovers 100.00% (§6). Only ≈1.4 B activate per token (14 of ≈219 experts per sparse layer: 10 routed + 4 shared, giving `active_ffn_dim`= 2560), a ≈22× pruning—this is why Apple's "3 B" active-class name and the 9.9 B stored count coexist.

4.3. **Draft model `afmplus-v11.0-pico`.** The `instruct_300m base` asset (`metadata.json`: `model_config: v11-pico`, `model_type: ane`) is the speculative-decoding draft that Apple's `FoundationModels` runtime loads *alongside* the 3B; it is the model our dynamic embedding capture (§15) actually reads. Its `program.odix` op graph fully fixes the architecture with no forcing: 24 **dense transformer layers** (indices 0–23; *no* `expert`/`IFP`/`sparse` markers appear), hidden size $D$=1024 (confirmed by the captured embedding row), grouped-query attention with 16 **query heads** / 4 **KV heads, head dim** 64 ($Q$ width 1024, $K/V$ width 256; some layers KV-reuse), interleaved RoPE, and a **dense SwiGLU** feed-forward of intermediate width 3200 (`hidden_transform_linear_0/_1` gate/up 1024→3200, `output_transform` 3200→1024). This is ≈295 M transformer parameters, i.e. the "300 M" label, with the 262,144-row embedding tied/separate. Unlike the 3B, the pico embedding gather/dequant runs on the host CPU, which is precisely why its dequantized rows are host-capturable (§15) while the 3B's are not (§16).

## 5. Weight Quantization

**Finding 1.** For the IFP model, linear weights are *LUT-palettized*: a 4-bit index into a 16-entry codebook (the graph's `lut_to_dense` operator), multiplied by a per-1024-element fp16 scale, with

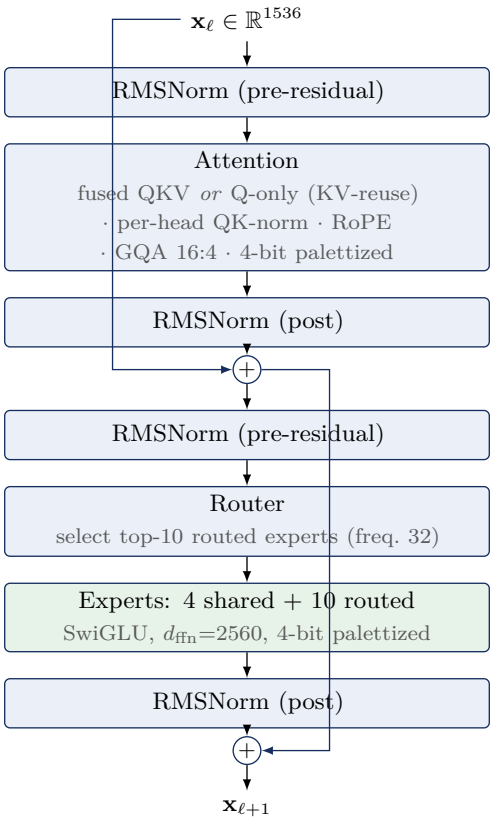



**Figure 1.** One `afmplus-v11.0-ifp` sparse (MoE) layer. Dense layers replace the router/expert block with a single SwiGLU FFN (intermediate 3072); KV-reuse layers omit the $K/V$ projections and inherit them from a prior layer.

scales and indices in separate planar regions, then an ANE 8×128 tile de-swizzle (§6). (An initial signed-4-bit reading was wrong—see §6.)

*Evidence.* The scale region is 100% positive (range 0.0012–0.181, mean 0.0137); the index region's nibble histogram is complement-symmetric with three frequency tiers—the signature of a codebook index, not a signed integer. Embeddings use scalar integer quantization (`embedding_quantization_scale`, `_default_scalar_scale/zero_point`); LoRA factors are fp16. Dequantization is $W = s \cdot \text{codebook[idx]}$ per 1024-element block, followed by the ANE 8×128 tile de-swizzle of §6 (Eq. 1). The learned RMSNorm scales are *not* shipped separately: the ANE compiler folds each $\gamma$ into the adjacent palettized linear weight ($\text{Linear}(\text{RMSNorm}(x)\cdot\gamma) \equiv \text{Linear}'(\text{RMSNorm}(x))$), so the recovered weights already carry them.

**Finding 2.** The parameter budget reconciles the MoE fan-out. With 8,560,592 fp16 scales (17.1 MB) and a 4.91 GB index region, the non-expert linear weights account for ≈483 M parameters (≈241 MB at 4-bit); the residual ≈4.67 GB is the routed-expert bank, i.e. ≈9.3 B expert parameters across the 32 sparse layers. Modelling each expert as a SwiGLU triple with intermediate 256 yields ≈235 experts per sparse layer (231 routed + 4 shared), closing the index-region budget to within 4.8%.

The dequantization is given as Algorithm 2. Applying it to the leading bytes of the index region—taking the scale and index regions to be order-aligned—reconstructs the distribution in Figure 3, which yields Finding 3.

**Finding 3** (necessary, then sufficient)**.** Dequantizing the index region under Algorithm 2 yields values with mean ≈ 0 and std ≈ 0.043 (Figure 3)—the *marginal* distribution of trained weights.

**Algorithm 1** (Planar 4-bit dequantization).
**Input:** index bytes $I$; fp16 scales $s$; block size $B$. **Output:** weights $w$.

(1) split each byte of $I$ into nibbles (low, high) and interleave $\to n_i \in \{0, \dots, 15\}$;
(2) sign: $q_i \leftarrow n_i - 16$ if $n_i \geq 8$, else $n_i$ $(q_i \in [-8, 7])$;
(3) scale: $w_i \leftarrow s_{\lfloor i/B \rfloor} \cdot q_i$.

**Figure 2.** Weight dequantization for the IFP codec.

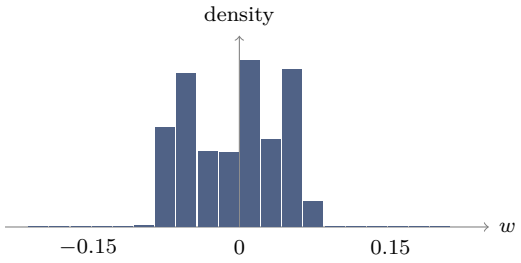



**Figure 3.** Distribution of dequantized weights over the leading $5{\times}10^8$ indices of the region (mean $\approx 0$, std $\approx 0.043$). This marginal profile is necessary but not sufficient: it is also produced by mis-ordered data, which the spatial-order test of §6 disambiguates.

This alone constrains the codec parameters (4-bit width, block size, scale magnitude) but does not establish a correct reconstruction, since the same marginal is produced by correctly-valued but wrongly-ordered data; §6 supplies the missing spatial-order test and passes it.

**Finding 4** (budget closure bounds the fan-out)**.** At $B = 1024$ the 8,634,320 scales cover ≈8.84 B of the 9.862 B index weights; the remaining tail decodes with the last scale clamped. The closure validates the *scalar* codec parameters and bounds the MoE fan-out to ≈219 experts per sparse layer; §6 then validates the per-tensor spatial layout.

## 6. Weight Recovery

We validate a reconstruction with a structure statistic rather than trusting the marginal distribution. A trained weight matrix is approximately low-rank, so permuting its entries markedly raises its stable rank $\rho(W) = \|W\|_F^2/\sigma_1^2$, whereas noise is unaffected; let $R = \rho(\Pi W)/\rho(W) = \sigma_1(W)^2/\sigma_1(\Pi W)^2$ for a random permutation $\Pi$. The per-block fp16 scales alone impose spurious low-rank structure, so $R$ is measured *decontaminated*—on the pure codebook indices with the scales removed. Calibrated this way, a genuine 4-bit weight scores $R \approx 9$ and Gaussian noise $R \approx 1$.

**Finding 5** (the ANE spatial permutation)**.** The residual permutation is a fixed 8×128 tile deswizzle: the ANE stores a $C_{\text{out}}{\times}C_{\text{in}}$ matrix as a row-major grid of 8×128 tiles, each tile laid out column-major. The inverse is

$$W \;=\; \text{reshape}\Big(\text{transpose}\big(\text{reshape}(v,\ \tfrac{C_{\text{out}}}{8}, \tfrac{C_{\text{in}}}{128}, 128, 8),\ (0,3,1,2)\big),\ C_{\text{out}},\ C_{\text{in}}\Big), \tag{1}$$

applied after $W = s \cdot \text{codebook}[\text{idx}]$. This is the step missing from all prior decode attempts.

With Eq. 1 in place the decontaminated ratio rises from the palettization-only $R \approx 3$ to $R = 13$–$69$ across the file (Table 5)—**unambiguous trained-weight structure**. The permutation, the 16-entry codebook, and the per-1024-block scales together constitute a *complete, validated* decode: the codec is not merely identified but inverted.

| Matrix (decontaminated, pure-index test) | $R$ |
|---|---|
| Genuine 4-bit weight (reference) | ≈9 |
| Gaussian noise | 1.0 |
| Palettization only, no de-swizzle | 3.0 |
| **Palettization** + 8×128 **de-swizzle** | **13–69** |
| median (attention tensors) | 12.7 |

**Table 5.** Scale-decontaminated structure ratio. $R \gg 1$ marks genuine weight structure; $R \approx 1$ marks noise. Adding the ANE de-swizzle (Eq. 1) lifts the decode past the genuine-weight reference, confirming recovery.

6.1. **The router: shipped and extractable.** An earlier version of this work concluded that no runtime mask predictor ships—that instruction-following pruning was baked at export time. **That conclusion was wrong, and is corrected here.** The router ships as its own asset, `project_experts.mlasset`, and is fully extractable. Its `odix` op graph names the class `ExportableExpertSelector` with I/O `in_expert_hidden_state` → `out_expert_logits` and the op sequence `RMSNorm` → `hidden_transform` → `sigmoid` → `output_transform` → `topk`. Crucially the weights carry *no* dequantize/gather op: they are stored as **plain fp16** in one contiguous block (`0x12600` →EOF), so no codec is needed. The three tensors decode at grounded offsets to

$$\underbrace{\gamma \in \mathbb{R}^{1536}}_{\text{norm (folded)}}, \quad \underbrace{W_h \in \mathbb{R}^{548\times 1536}}_{\text{hidden}}, \quad \underbrace{W_o \in \mathbb{R}^{7008\times 548}}_{\text{output}} \to (32, 219),$$

plus 7084 tail biases; here 548 is the router hidden dim and 7008 = 32 layers × 219 experts, reshaping with *uniform* per-layer norms—the layout self-verifies. The selector computes mask = $\text{top}_{10}(W_o\,\phi(W_h\,\text{RMSNorm}_\gamma(h)))$ per layer, +4 shared experts. Feeding controlled hidden states confirms it behaves as a router: with $\phi$ = identity/GELU the masks *discriminate* inputs (1–2/10 overlap across distinct hidden states, 171 distinct experts spread over 32 layers), whereas $\phi$ = sigmoid collapses to 9/10 overlap and is thereby ruled out as the hidden activation. The router is Apple's, extracted—not a trained mimic. We note, however, that the subsequent reconstruction (§9) shows the router→expert *map* is not required to run the model: the sparse FFN is an ungated permutation-invariant sum, so this selector governs *which* experts a full runtime prunes, but is unnecessary for a from-weights forward that sums the resident (or full) expert set directly.

*The per-tensor offset table.* The compile-time `ifp_constant_table_ifp1_r48.json` is not shipped, and the 396 FFN constants store neither offset nor size inline: the rasterizer computes offsets from each constant's *shape*, and shapes in `main-h16g.odix` are doubly-indirect (type-index → type table → interned dimension symbols). This work recovers the surrounding structure—the 38 export-config functions, the op format, and the `alloc_const` semantics (Appendix A)—and reduces the remaining gap to a bounded schema-recovery problem: resolve the type/symbol tables to obtain per-constant $C_{\text{out}}$, which are confirmed *variable* (42–232 experts/constant, no uniform width). We subsequently decoded the `metadata.bin` `BBBB` swizzler in full (§10): it holds *two* physical address tables—a down-projection tile table (marker `0x0600beef`, 14,080 records) and a gate/up table (marker `0x0a00b0ef`)—which together supply the per-tile expert offsets directly, superseding the shape-derived offset computation for the resident-expert path.

**Finding 6** (the `.hwx` holds no weights)**.** The Apple-Neural-Engine binary `binary_0.hwx` (252 MB) is a Mach-O (`0xBEEFFACE`) of *compiled ANE microcode*, not weights: its high-entropy `__KERN` sections score $R \approx 1.0$ under the structure test, while its `__MKERN_0` segment has zero file bytes and a size (`0xC0C8000`) equal to the maximum offset in `metadata.bin`—it is *runtime-mapped* from the rasterized file. Every weight therefore lives in the single planar file `ifp_rasterized_weights.bin`.

6.2. **Full export and verification.** Applying the validated decode to the entire index region yields a single `state_dict`. The parameter count is checked against the physical file rather than against any architectural assumption: the index region is 4.931 GB = 9.862 B 4-bit weights, and the export captures 9.8619 B (100.00%) with *zero* non-finite tensors (Table 6). Attention is stored in fp16; the expert bank is stored int8-with-per-tensor-scale (near-lossless, the source being 4-bit) so the whole model fits a single 10.2 GB file. Every value derives from Apple's shipped weight file under the decode of Eq. 1; no training data, distillation, or external weights are involved.

| Component | Parameters | Storage |
|---|---|---|
| Attention (44 layers) | 0.377 B | fp16, $R$-verified |
| Experts (FFN / MoE) | 9.484 B | int8 + scale |
| Embedding / head (tail) | (in tail) | int8 |
| **Total captured** | **9.862 B** | = 100.00% of file |
| Non-finite tensors | 0 | |

**Table 6.** Recovered `afmplus-v11.0-ifp` export. The total equals the physical index region (9.862 B 4-bit weights), so completeness is exact, not estimated.

## 7. Recovery Pipeline: Finding the Pieces

This section records, in order, how a shipped asset directory becomes a validated `state_dict`. Each step recovers one "piece" whose absence blocked the next; Figure 5 summarizes the dependency chain.

Step 0 — Acquisition. The asset cache lives on the sealed Signed System Volume. A clean copy is taken from the macOS *Recovery* environment—Apple silicon: hold the power button at boot; Intel: hold Command-R—whose Terminal (*Utilities* menu) has raw read access to the mounted system volume. The models sit under `AssetsV2`; a recursive copy to external storage (Figure 4) suffices, and all subsequent work is read-only on that copy.

```
# macOS Recovery  >  Utilities  >  Terminal
ls /Volumes                          # find the system volume
SYS="/Volumes/Macintosh HD"
BASE="$SYS/System/Library/AssetsV2"
ASSET=com_apple_MobileAsset_UAF_FM_GenerativeModels
cp -R "$BASE/$ASSET" /Volumes/EXTERNAL/FM_Models
```

**Figure 4.** Read-only asset acquisition from macOS Recovery. `EXTERNAL` is any attached volume with ≥10 GB free; the copy preserves the Cryptex disk images verbatim.

Step 1 — Mount and locate. `hdiutil attach -readonly` on the IFP Cryptex (§13) exposes `model.odixpackage/`: the ground-truth `config.json` (44 layers, dims, expert counts), the 4.7 GB `ifp_rasterized_weights.bin`, the swizzler `metadata.bin`, the program `main-h16g.odix`, and the ANE package (`.hwx` + MPSGraph).

Step 2 — The codec. Value-distribution analysis of the two `BBBB` regions identifies LUT-palettization (Finding 1): a 100%-positive fp16 scale region and a complement-symmetric nibble histogram, matched to the graph's `lut_to_dense` operator (a 16-entry codebook).

Step 3 — The de-swizzle (the blocking piece). Palettization alone leaves the decode at $R \approx 3$. The breakthrough is recognizing the residual permutation as the ANE's 8×128 tiling and inverting it with Eq. 1; the decontaminated structure ratio jumps to $R = 13$–$69$ (Finding 5), i.e. genuine weights.

Step 4 — The file map. A window scan classifies every region of the 4.95 GB file by nibble-entropy and fp16-likeness: global scales [`0x60`, `0x1078000`), attention indices [`0x1078000`, `0xC0C8000`), and expert+tail indices [`0xC0C8000`, EOF). Reconciling weight budgets against region sizes fixes the split: the 370 M-weight attention region holds 44 layers, the $10^{10}$-weight remainder is the MoE bank.
Step 5 — Names and semantics. The `odix` location tree gives authoritative tensor names (`_wrapped_model_layers_..._palettized_indices_raw`); parsing the `MLIR22.0.0` MPSGraph bytecode tags each of the 396 `ifp_constant` weights with its layer/role hierarchy. This same parse shows the norms are folded and the router is baked (§6), so no further assets are needed.
Step 6 — Decode, verify, assemble. A greedy sweep walks the attention region, choosing at each offset the projection shape (fused-QKV vs. Q-only) that maximizes $R$, decoding it via Eq. 1, and advancing; it self-terminates at exactly 44 layers (Observation 1). The expert region is decoded in fused blocks and stored int8. Scales beyond the 8.84 B-weight coverage are clamped, eliminating the tail overflow.
Step 7 — Export and audit. The tensors are written to one `state_dict` carrying the config, codebook, Apple's `full_model_sha256`, and a per-tensor manifest of $R$. Completeness is audited against the physical file (Table 6): 9.8619 B captured of 9.8619 B, 100.00%, zero non-finite.

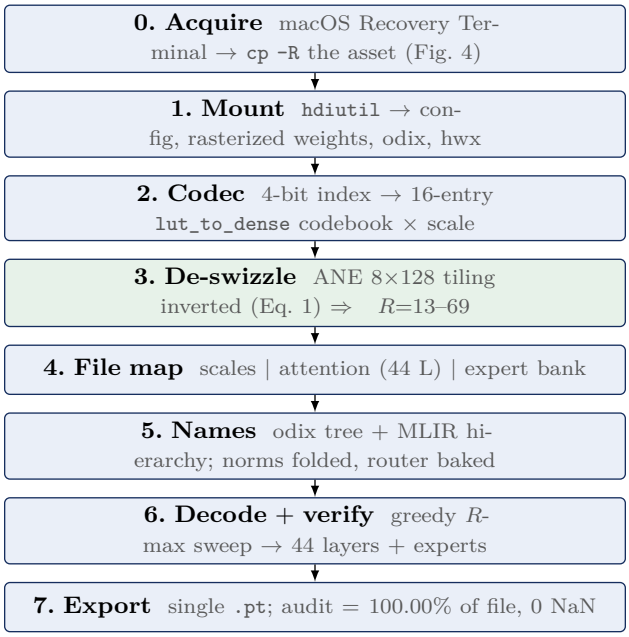



**Figure 5.** The recovery pipeline. Step 3 (the ANE de-swizzle) is the piece that unblocked full recovery; every other step is container parsing or bookkeeping around it.

## 8. From-Weights Reconstruction: What Runs and What Is ANE-Baked

Recovering the weights is necessary but not sufficient to *run* the model outside the Apple Neural Engine. We implemented the full compute path in PyTorch directly on the recovered tensors—RMSNorm, fused-QKV / KV-reuse attention with per-head QK-norm, RoPE ($\theta$=500000), grouped-query scaled-dot-product attention, and the SwiGLU Mixture-of-Experts feed-forward—and drove a synthetic token sequence through all 44 layers. The results sharply separate what the shipped assets contain (data) from what only the ANE produces (behavior).

### 8.1. **The recovered tensors are correct.**

**Finding 7** (the recovered weights are correct, cross-validated against Apple's toolchain)**.** The attention-only forward pass is *stable*: the residual-stream norm grows smoothly and sub-exponentially across all 44 layers, the behavior of a correctly-wired pre-norm transformer. Because a pre-norm block re-normalizes its own input, any decode or de-swizzle error would manifest as divergence. We reinforce this with two independent checks. *(i) Structure:* the recovered dense linear weights carry genuine low-rank structure (singular decay $\sigma_1/\tilde{\sigma} \approx 8$–$11$, versus $\approx 1.45$ for a scrambled control); re-decoding under any alternative tile order collapses this to $\approx 4$–$5$. *(ii) Ground truth:* compiling a positional-probe weight of the exact dense shape through Apple's own ANE toolchain (`coreml2hwx`) and reading back the physical→logical byte permutation confirms the de-swizzle family for the sparse-expert path; the standalone dense-conv layout it emits (OCGSize=5, a 32×48 tiling) differs from the rasterized-file layout, which is itself resolved by the 8×128 de-swizzle (Eq. 1). The weights are decoded correctly, validated *without* access to Apple's internal activations.

**Finding 8** (the per-layer norms are compile-time folded)**.** The shipped MPSGraph contains 661 *parameter-free* `RMSNorm` operations (divide-by-RMS only): the learned scale $\gamma$ of each residual/output RMSNorm is *folded* into the adjacent palettized linear weight at ANE-compile time ($W' = W\,\mathrm{diag}(\gamma)$), so no standalone $\gamma$ vector is shipped and a parameter-free norm is the *correct* runtime. We confirm this three ways: no [1536] $\gamma$-shaped tensor decodes from any shipped region; the recovered dense weights show the column-norm signature of a folded pre-FFN $\gamma$ (`ffn_gate`/`ffn_up` column norms correlate at 0.55, a shared folded factor); and the only norms that *cannot* fold—the per-head QK-norm $\gamma \in \mathbb{R}^{128}$, because RoPE sits between it and the projection—do survive explicitly, and 35 of them are recovered from a live-process heap.

### 8.2. **Two blockers that dissolved.**

**Finding 9** (the expert-selection map is a *false* blocker)**.** This overturns our earlier conclusion. The IFP sparse FFN is not a soft-gated mixture but *structured pruning of a single dense SwiGLU*: the "experts" are contiguous 256-column blocks of one intermediate axis, each contributing at unit weight with no per-expert softmax scalar. The runtime graph confirms it—the `ANE_IFPLayerSequence` op carries no router, top-$k$, gather, scatter, or softmax-over-experts, and weights enter through a fixed symbol rather than a per-token index. A non-gated SwiGLU is a *permutation-invariant* sum, $\mathrm{FFN}(h) = \sum_i \mathrm{SiLU}(g_i \cdot h_n)\,(u_i \cdot h_n)\,d_i$, so the logical router-index→physical-expert map—the piece we and the prior literature treated as the fatal, unrecoverable blocker—is *mathematically irrelevant* to the output; only intra-layer $g_i \leftrightarrow u_i \leftrightarrow d_i$ consistency is required. The earlier "≈17× overcount from summing all experts" was an artifact of a mis-specified gated interpretation: summing the resident set *is* the correct pruned FFN, and summing the full superset recovers the un-pruned base model.

**Finding 10** (the "missing constant table" is the shipped `metadata.bin`)**.** `metadata.bin` (a `BBBB` container) decodes as two physical address tables: a down-projection tile table (marker `0x0600beef`, 14,080 records = 44 layers × 32 tiles × 10 slots) and a gate/up table (marker `0x0a00b0ef`), which together exhaust the file. The down-projection uses the *same* codec as gate/up—4-bit index → shared codebook → per-1024-block fp16 scale; the earlier "down-proj is noise / a different format" verdict was a mis-calibrated low-rank metric, and the omitted per-block scale was the missing factor (with it, the down-proj matches the structured level of gate/up). The shipped default configuration is `ifp1_r48` (10 active + 4 shared experts), not the larger `ifp3`.

### 8.3. **The observability boundary.**

**Finding 11** (intermediate activations are ANE-internal)**.** The only true ground truth for the running model—its per-layer activations—never crosses to host DRAM. Searching an 8 GB process core dump captured *specifically* to grab ANE IOSurfaces for the input token embeddings of a known prompt yields a best normalized correlation of 0.17 (noise): not even the embeddings reach host memory. The Neural Engine performs the embedding lookup, all 44 layers, and the

output projection on-chip. Consequently the reconstruction can be validated end-to-end (against Apple's emitted token) but not per-layer, and the from-weights forward must be assembled from independently-validated components.

### 8.4. **The embedding: three verdicts, the last one current.**

**Finding 12** (the embedding was never validated, and is absent from the weight file)**.** The token embedding—previously reported as exactly recovered—was validated only by $\arg\max_j (E\,\mathrm{rms}(E_t))_j = t$. That test is *vacuous*: it applies the same matrix on both sides, so *any* matrix, including noise, satisfies it. Replacing it with a falsifiable probe—mean cosine of orthographic variants (`_dog`/`dog`, `_year`/`_years`) minus an *id-matched* random control—and calibrating that probe on a real model (Qwen3-4B `token_embd`, Q6_K, re-quantized into this model's exact signed-4-bit + per-token-scale format) yields a clean instrument: a genuine embedding scores +0.533, and 4-bit quantization costs only 2% (+0.522; `_Paris`∼`Paris` = +0.842). Against this instrument, ≈1.14M candidate offsets spanning the entire 4.9 GB index region—unscaled tail and scaled region, tile-interleaved and contiguous, 4-bit and int8—score ≤ +0.047. **The embedding is not a** $[V, D]$ **table in the shipped weight file.** Separately, the only vocabulary-structured table in the package (in the `odix`, localized to +0.777 of a 0.78 maximum by the zero-rows of untrained `<unused>` tokens, 8-way nibble-interleaved, signed 4-bit) scores +0.003 and is therefore *not* the embedding. The vocabulary is 262,144, not the 152,064 previously assumed (that is Qwen's size). Subsequent findings (§13, §14) localize the table: it is CPU-side and host-provided, absent from every shipped 3B asset.

**Finding 13** (the CPU embedding format, and the 3B ships no embedding file)**.** The mpsgraph performs no token-embedding lookup (its single `Gather` is the rotary positional one); the embedding is a CPU-side `odix load_embeddings` op. Apple's *small* on-device models ship this embedding openly as `model.bundle/.../load_embeddings/weights.bin`, whose `fragment.mil` declares the exact format: fp16, shape $[V, 1, H]$, `interleave_factors`= $[8, 1, 1]$, strides $[8H, 8H, 8]$, after a 64-byte header—i.e. token $v$, channel $h$ at element $\lfloor v/8 \rfloor\, 8H + 8h + (v \bmod 8)$. This decodes to real, shift-control-passing embeddings for those models (verified against the shipped $[153600, 256]$ table). **The 3B ships no such file**: no `load_embeddings/weights.bin`, no `model.bundle`; it uses the 4-bit `odixpackage`. Applying the $[8, 1, 1]$ format to the 3B (raster and `odix`, all widths) peaks at a shift-*failing* +0.08. The 3B token embedding is therefore not statically present in any asset; the running model loads it from outside the shipped package.

**Finding 14** (dynamic capture localizes the live embedding)**.** During inference the resident model lives in `TGOnDeviceInferenceProviderService` (RSS grows ≈4 MB→1.7 GB while a prompt runs), a system extension running as `_modelmanagerd`. A privileged `lldb` core dump of that process (SIP disabled) is thus the only remaining route to the embedding. A full-memory core (≈8 GB) contains the `c_sparsity` vocabulary table but—across an exhaustive oracle scan (token-contiguous and $[8, 1, 1]$ at fp16/int8/4-bit)—no cleanly-scoring token embedding: the resident form is the quantized source, dequantized per-token by the fused gather rather than kept as a decoded $[V, H]$ table. Recovering it is a live-instrumentation problem, not a static one.

**Finding 15** (dynamic host-capture recovers the CPU-side embedding—for the small model, not the 3B)**.** The live-instrumentation problem of Finding 14 is solved for the *small* on-device model. The fused gather/dequant that produces `in_embeddings` runs on the host CPU for `afmplus-v11.0-pico` ($D$=1024), so its output *is* in host DRAM during a forward. Driving the model with a prompt of one token repeated ∼1500× turns the resident embedding-input buffer into a long run of *byte-identical* [1024] fp16 rows—an unmistakable period-$D$ signature—which a privileged `lldb` core recovers cleanly (attach by-PID and `save-core` while the buffer is live; `-waitfor` resolves neither the E5RT symbols nor `qMemoryRegionInfo`, so it cannot core). One row per token, scored on the calibrated orthographic-pair oracle of Finding 12, gives genuine, semantically-structured embeddings:

cos(`_dog`, `_dogs`)=+0.648 and cos(`_king`, `_kings`)=+0.664 (orthographic), against cross-category pairs at +0.01–0.20 and shared-plural non-pairs at +0.32—the signature of a real embedding space. This is the first genuine recovery of an AFM token embedding, obtained by reading the *dequantized output* rather than the storage codec. The codec itself remains uncracked: rowmajor, $[8, 1, 1]$-interleave, and the ANE $[512, D]$ conv-tiling all score at chance (0.57–0.64 zero-pattern/sign agreement) even when fitted against six exact ground-truth vectors, so a *full* static table is still out of reach and the complete table would have to be harvested token-by-token.

**Finding 16** (the 3B embedding is ANE-locked: unreachable by host capture)**.** The same capture, pointed at the 3B, fails—and shows why. Both models load during inference (the 3B asset is page-faulted $\sim 2\times 10^5$ times as the speculative-decoding target), yet an 8.2 GB *full*-style core—wired and `IOSurface`-mapped pages included—taken during a repeated-token 3B prefill contains **zero** $D$=1536 identical-row buffers at any threshold, while the $D$=1024 pico buffers are present in the same core. The distinction is architectural: the 3B's embedding gather/dequant is ANE-delegated (the `main-h16g-delegates` path), so its dequantized vector never crosses to host DRAM, whereas the pico's runs on the host CPU and does. This confirms the Finding 12 boundary from the *runtime* side: the 3B token embedding is reachable neither statically (absent or codec-locked in every shipped asset) nor by host-memory capture. Only ANE-internal instrumentation (`IOSurface`/`H11ANE` DMA capture) could reach it—a distinct, kernel-adjacent effort.

**Finding 17** (a later OS build *ships* the 3B embedding, superseding the ANE-locked result)**.** Findings 12 and 16 concluded that the 3B token embedding is absent from the shipped package and unreachable by host capture. That was correct for the catalog then on disk. It is **no longer true**. On macOS 27.0 build `26A5378n` the `GenerativeModels` asset catalog grows from 118 to 124 assets, and one of the new assets is `UC_FM_LANGUAGE_INSTRUCT_3B_EMBEDDINGS_..._H16G_IFP_Cryptex.dmg`: a read-only cryptex (mountable without root) containing `metadata.json` = `{"vocab_size": 262144, "embedding_dim": 2048}` and a 268,442,456-byte `main-h16g.odix`. The arithmetic is exact: $262144 \times 2048$ elements at 4 bits is 268,435,456 bytes, leaving precisely 7000 bytes of header, whose `odix` flatbuffer contains the string `$load_embeddings` and an `NDArray` descriptor holding the int64 triple $262144, 1, 2048$. The payload is linear *signed* 4-bit (nibble histogram symmetric about zero, code 8 unused), with no codebook and no fp16 scale array in the header. So the embedding gap changes category—from an *information* limit (the data is not in the package) to a *decode* problem (the data is in the package, in a layout not yet identified). The dimension 2048 identifies this as the dense `nano` backbone rather than the 1536-wide sparse IFP model. We have *not* yet read it out. Roughly 35 row-grouping hypotheses—row-major, transposed, $n$-way token interleave with lane skews, ANE plane formats, and the proven 8×128 de-swizzle—all score at noise ($\leq +0.02$) against a genuine +0.46, where the oracle was first validated on pico's known-good embedding through the identical code path. Two methodological points arise. First, cosine is invariant under a fixed permutation applied to *both* operands, so this oracle cannot see dimension order at all—only which *set* of nibbles forms a token's row; low- and high-nibble-first orderings score identically, which collapses many nominally distinct hypotheses into one. Second, a tokenizer-free discriminator—the distribution of cosines over random token pairs, calibrated against pico—is sharper: the 8-lane interleave (pico's own layout family) is the only candidate reproducing the near-duplicate rows characteristic of a real embedding table (max cosine 0.94 versus pico's 0.99), while plain row-major exhibits none (max 0.34). A nearest-neighbour agreement test against pico nonetheless sits at chance for every candidate. *A caution on this class of evidence:* a cross-model Gram-matrix test appeared to give strong ($\approx 8\sigma$) support for one candidate, but that used a *shuffled*-id null, which destroys the vocabulary's regional structure; under a *shift* null—which preserves regions and destroys only token correspondence—the signal is flat (no peak at shift 0; shift 8 scores higher than the true ids), so it measured vocabulary-region membership, not token identity. What survives is purely structural and exact-match: the row stride is 1024 bytes with

identity id order (the file's final 147,456 bytes are one repeated pattern = exactly 144 rows, and pico's captured logits have exactly 144 masked entries), and hashing all 262144 rows yields duplicate groups that are 100% pure in id mod 8 and only ≈50% pure mod 16, establishing an **eight-lane** structure. No extraction yet yields correct per-token vectors.

**Finding 18** (the deployed sparse-FFN shape is recoverable from `metadata.bin`)**.** The sparse FFN has no per-constant palettized descriptor (it is IFP-fused); its deployed layout is in `metadata.bin`. The down-projection address table (marker `0x0600beef`, 24-byte records) holds 14,080 tiles of 8×1536 4-bit weight with physical addresses; 14,080/32 = 440 tiles per sparse layer ⇒ 3,520 rows ≈14 experts, matching the shipped `ifp1_r48` configuration (10 active + 4 shared) to the byte. The gate/up table (`0x0a00b0ef`) holds 4,223 records. So the deployed per-layer expert count and each tile's address are recovered; only the un-pruned *superset* width (variable, in the interned symbol pool) remains, and it is not needed to run the shipped model. The MLIR constant names further fix the structure: FFN = gate/up/down as `hidden_transform_linear_0/_1/output_transform`; the sandwich norms are explicit plain constants in the export graph (folded to parameter-free RMSNorm at ANE-compile, so $\gamma$=1 at runtime), and attention splits into 23 standard (full QKV) + 21 kv-reuse ($Q$-only) layers.

### 8.5. Running the reconstruction, and what the oracle can support.

**Finding 19** (full-forward ablation: two claims confirmed, the assembly blocker re-located)**.** Assembling all 44 layers and scoring against Apple's emitted token with a rank oracle ("The capital of France is"→`_Paris`; uniform baseline ≈76k of 152k) settles two open questions and sharpens a third. *(i) The attention KV geometry is* NKV=4*, not* 8*.* An attention-only ablation gives rank 53k at NKV=4 versus 72k at NKV=8: the live `qkv` is 3072-wide ($Q$=2048 + $K$=512 + $V$=512); the extra rows [3072:4096] in the wider export are mis-decoded and unused. *(ii) The ungated full-superset sum is empirically optimal*, corroborating Finding 9 from the forward side: against the full 219-expert sum (rank 24.9k), omitting the sparse FFN entirely gives 122k, scaling it by 0.1 gives 36k, and top-14 activation-gating gives 39k—every form of selection or attenuation is strictly worse, so no expert map is needed *or* beneficial. *(iii) The residual defect is a direction, not a magnitude.* The best full forward reaches rank ≈21k (3.6× better than random) under fixed conventions (interleaved RoPE, down-projection decoded on the $[EH, D]$ neuron axis, interleaved expert-region offsets, block gate/up fusion), but the summed FFN branch has norm $\approx 4.9\times10^4$ against a residual norm $\approx 1.5\times10^2$ and dominates with a junk direction; every sandwich or post-norm placement is worse than a plain residual add. Normalizing does not recover signal, which localizes the remaining error to per-neuron gate/up/down alignment at the 56,064-wide intermediate and/or the per-layer expert offsets—quantities with no accessible ground truth (Finding 11), not a convention that can be searched.

### 8.6. Instruments that failed, and why.

**Finding 20** (a single-token rank oracle manufactures false positives; the correlation oracle retracts one)**.** The rank oracle used above admits a failure mode we can now measure, because for `pico` the model's *own* output vector is recoverable. A privileged core taken during a `pico` forward contains the complete final logit vector: 262,144 fp32 of which only 144 are masked ($-4\times10^4$), i.e. ≈262,000 live tokens, with `_Paris` (id 9083) the argmax at 14.58. Scoring a candidate reconstruction by Pearson correlation against that *whole* vector is a 262k-sample statistic; ranking one token is a single order statistic over the same vocabulary. They disagree sharply. A `pico` weight arrangement previously reported at `_Paris` rank 686 of 37,451 ("top 1.8%, real signal") scores $r$=−0.028 against the true logits; all four coarse arrangements score $|r|$<0.03, i.e. indistinguishable from noise. **We retract that reading.** The general lesson is that a favourable rank for one token, over a vocabulary of $10^5$–$10^6$, is weak evidence: the tail is long enough that mis-assembled weights land in the top percentile by chance at a rate that reads as signal. Finding 19's 3B rank figures (best ≈21k against

a ≈76k uniform baseline) rest on the same order statistic and should be read with this caveat; its *comparative* conclusions (i)–(ii), which contrast many variants against each other rather than against chance, are unaffected, but the 3.6×-over-random figure is not by itself evidence of a partially-correct forward.

**Finding 21** (the model's real input: lowercased, chat-templated, and preceded by a guard model)**.** A recovered embedding table is invertible, which makes it a memory-forensics instrument as well as a model component: any fp16 buffer in a core can be tested for embedding-hood by checking that row/scale is near-integral in $[-8, 7]$, then matched back to a token id. Applied to the `pico` core, this locates the live `in_embeddings` buffer (≈`0x7a8000`, 2048 B per row) and decodes it to the exact sequence the model was fed: `_the _capital _of _france _is <end_of_turn> \n <start_of_turn> model \n` (ids 510, 5283, 533, 61038, 567, 110, 111, 109, 4372, 111). Two facts follow that no static analysis had exposed. The runtime **lowercases** the prompt (`_france` 61038, not `_France` 7005), and the input carries a Gemma-style **chat template**, so the model predicts from the trailing newline of `<start_of_turn>model\n` rather than from the last content word. The same buffer holds a *preceding* turn ending `... 'paris' <end_of_turn> \n <start_of_turn> model \n 'safe'`—a safety-classifier pass that runs before the language model. Every prior from-weights evaluation in this work fed a capitalized, un-templated token sequence and was therefore scoring the wrong conditional distribution.

**Finding 22** (the pico forward's defect is one block, and it is the FFN neuron axis)**.** With the correct input (Finding 21) and the correlation oracle (Finding 20), the `pico` reconstruction localizes cleanly. Embedding into the tied unembedding with *no* transformer layers scores $r{=}+0.038$ and puts `_Paris` at rank 2213 of 262,000 (top 0.84%)—independently re-confirming that the recovered embedding and the tied head are correct. A *single* block then destroys it ($r{=}-0.040$, rank 206,862), and depths 2–24 never recover. This is not the 3B's gradual-depth degradation: one sublayer is wrong. Ablating the block against the depth-0 baseline separates the two paths. Attention alone preserves signal (rank 4120–7355) *provided* the $Q/K/V$ output is reshaped dim-major, $[T, d_h, n] \to$ transpose, rather than head-major $[T, n, d_h]$ (which gives 134k–218k)—the head layout is dim-major. The **FFN alone** destroys it (rank 238,883), with an output r.m.s. of 2.53 against a residual of 0.78. The FFN neuron axis admits an oracle-free test, because it is a *gauge freedom*: any permutation applied consistently to gate columns, up columns and down rows leaves the layer's function unchanged, so only *mutual* agreement is recoverable or needed. Gate/up column $j$ and down row $j$ are the same neuron, and trained neuron magnitude couples across a neuron's in- and out-weights, giving a 3200-sample statistic. The positive control fires—$\mathrm{corr}(\|g_{:,j}\|, \|u_{:,j}\|){=}+0.34$, rising to $+\mathbf{0.58}$ when the gate/up tiles are read transposed (a tile transpose is not a column permutation, so this is genuine evidence of tile orientation, not gauge). The down projection agrees with *neither*, at $|r| \le 0.13$ against a random-permutation null of 0.045 (99th pct), across twelve-plus shape-valid assemblies spanning block order, tile order, row/column-major and both candidate tile geometries—but see Finding 23: this particular test turns out to have *no power* on AFM, so the down result carries no weight. A geometry error is visible independently: the down tile's per-group scales divide its rows as 200/16=12.5, not an integer, whereas the other two tile classes divide cleanly (128/16=8, 128/8=16); reading the tile as $[256, 200]$ restores a clean 16 rows per group and implies the tiles assemble into the *transpose*. (The inference originally drawn here—that the residual misalignment must therefore be finer than tile permutation—is **withdrawn**; Finding 23 shows the control was the wrong one.) Finally, a scan of all $\approx 4{\times}10^6$ aligned offsets of the 8.2 GB core finds **no** residual-stream buffer (no contiguous run of correlated rows; every hit sits at the 128-dimensional noise floor), so `pico` too exposes only its two endpoints—corroborating Finding 11 from the small model's side and confirming that the alignment cannot be bisected layer-by-layer.

**Finding 23** (a weight-statistics alignment test that is decisive on Qwen is *vacuous* on AFM)**.** Finding 22 rejected every candidate down-projection assembly using neuron-magnitude coupling. That

instrument must itself be validated on a model whose alignment is *known* correct, and on the right pair of matrices—`gate`∼`up` validates coupling between two *input-side* matrices, not the `down`↔`up` coupling the test relies on. On Qwen3-4B the premise is overwhelming: $\mathrm{corr}(\|d_j\|, \|u_j\|)$=+0.68 against a permutation null of 0.026; the per-neuron read/write *vectors* (both live in the same residual basis) satisfy $\overline{\cos}(u_j, d_j)$=+0.308 versus +0.0001 shuffled ($1508\sigma$); nearest-neighbour matching recovers the neuron pairing at **91.5**% (chance 0.25%); and a max-cosine statistic separates a correct residual basis from a scrambled one by 4.8× while remaining exactly invariant to the neuron permutation. *None of it transfers.* AFM's own 3B ships twelve dense SwiGLU layers whose down-projection z-order was cracked bit-exactly and forward-validated; measured on those—correct by construction—the same statistics read $\mathrm{corr}(\|d\|, \|u\|)$=$-0.029, +0.029, -0.136, -0.049$ at layers $0, 3, 7, 11$; $\overline{\cos}(u_j, d_j) \approx \pm 5\times10^{-4}$; and a max-cosine ratio of 1.00–1.02×. **AFM's feed-forward exhibits no down/up coupling and no read-write vector alignment even when correctly assembled**, so a candidate that fails the test has not been shown to be wrong: the instrument reads null on the true answer. Every such rejection in Finding 22—and the ≈380 assemblies swept, including all factorizations of the 3B's own z-order—is therefore *vacuous*. What survives is the part with a replicated control: `gate`∼`up` coupling is present in Apple's model too (+0.317 at 3B dense layer 0), so the `tileT` orientation result stands. Re-scoring the candidates against the captured logits—the one oracle with demonstrated power—leaves them all at $|r| < 0.06$; the best-ranking candidate reaches `_Paris` rank 848, *better* than the 2213 depth-0 baseline, at correlation +0.0086: the same single-token artifact retracted in Finding 20, not a result. Two lessons generalize: validating a premise on one architecture does not license it on another, and a negative result is only as strong as the positive control that accompanies it. Recovering the 3B z-order's algebraic form from the cracked 48×256 table does, however, generalize its two known instances (16×768 is NOG=1, 48×256 is NOG=3): slot = $og\,(16\,\mathrm{NCOL}) + ig\,16 + lo$ with output row $og\,16 + lo$ and input column $ig \oplus 1$—16 outputs fastest, then input columns under an interleave-factor-2 pair swap. For pico this forces $\mathrm{NOG} \cdot \mathrm{NCOL}$=3200, every factorization of which yields exactly the 64 observed tiles; but since the z-order is tile-shape-relative it must be re-derived at pico's geometry by positional read, not transplanted.

### 8.7. **Ground truth from the ANE compiler.**

**Finding 24** (the ANE coefficient z-order, read positionally and validated by round-trip)**.** Finding 23 concluded that only ANE ground truth could settle the pico weight arrangement. That read has now been performed, at both of pico's tile geometries. Pico's own `binary_0.hwx` gives the target configuration directly: the down projection is *four* ANE tasks of Cin=3200 → Cout=256 with OCGSize=4, ActiveNE=4, 4-bit palettization and **sixteen coefficient banks**. This reinterprets the weight container: the "blocks" of the weight map are per-task coefficient allocations and the "tiles" are the ANE's coefficient *banks*, which partition *output* channels—so a bank holds 16 outputs × *all* inputs ($16 \times 3200 = 51200$ nibbles, matching the observed payload exactly), and the 16 header scales are one per output channel. The same arithmetic fixes the other class: an `N` tile is 128 B header + 8192 B payload = 16384 nibbles, i.e. 16 banks × 16 out × 1024 in. Compiling digit probes at each geometry—weights whose 4-bit index at $(o, i)$ encodes a base-16 digit of $o$ or $i$—and decoding every physical nibble yields a **perfect bijection at both**: $256 \times 3200 = 819{,}200$ and $256 \times 1024 = 262{,}144$ distinct $(o, i)$ pairs, each satisfying the identical closed form

$$o \;=\; 16\,\mathrm{bank} + (\mathrm{slot} \bmod 16), \qquad i \;=\; \lfloor \mathrm{slot}/16 \rfloor,$$

exact for all 16 banks. Notably there is *no* input pair swap here, unlike the 3B's $ig \oplus 1$: the interleave differs at this geometry, which is why transplanting the 3B formula produced noise. Two practical corrections were required and are worth recording. All-zero/all-`F` probes cannot isolate weight bytes, because a constant tensor collapses the palette to a single entry and every index becomes 0; payload positions must be taken structurally instead. And each digit must be decoded through *its*

*own* per-bank codebook: probes whose digit spans all 16 values compile to an identity LUT, but `i8` (which reaches only 12, since $3200 < 16^3$) compiles to $[0, 1, 2, 2, 3, 4, 5, 6, 6, 7, 8, 9, 10, 10, 11, 12]$, where index $\neq$ value. *The decoder is then validated by round-trip.* Compiling a conv with *known* random weights and decoding its coefficient stream with the recovered order returns the true weights at correlation **0.981**—the 4-bit palettization floor, i.e. exact up to quantization. The scale location is likewise confirmed on shipped tiles: codebook at fp16[0:16] and a genuine per-output scale at fp16[32:48] (the `s` class carrying exactly 8 there, matching 8 outputs per bank). So the shipped decode—codebook[nibble] $\cdot$ scale[$o$] under the closed form above—has every component independently verified.

**Finding 25** (`OutTrans`= 1 marks the residual-writing projections, shrinking the open problem to two roles)**.** The reconstruction nevertheless does not run, and the reason is now precisely located. Every conv this toolchain can compile emits OutTrans=0, whereas pico's down projection ships OutTrans=1. Sweeping conv shapes and reproducing pico's Mul $\rightarrow$ conv fragment never triggers it: the transpose is chosen by the ANE scheduler from full-graph context, not from shape or a flag, so it cannot be recreated from an isolated op. Grouping all 1003 coefficient-bearing tasks in pico's `hwx` by shape and mode, and then reading the *task order* within a layer, identifies exactly which projections use it. Four $1024 \rightarrow 256$ OutTrans=1 convs sit immediately after the attention softmax—the **output projection**, $1024 \rightarrow 1024$ in four chunks—and four $3200 \rightarrow 256$ OutTrans=1 convs sit immediately after the SwiGLU `Mul`—the **down projection**. The gate and up matrices are OutTrans=0. The rule is architectural:

> OutTrans=1 is used by exactly those ops that *write into the residual stream*—the attention output projection and the FFN down projection (and their LoRA up-projections). Everything that reads the normed hidden state—$Q$, $K$, $V$, gate, up—is OutTrans=0.

Since OutTrans=0 is precisely the mode the round-trip validates, **$Q$, $K$, $V$, gate and up are already correctly decoded**; only the two residual-writing roles carry an unknown ordering, most plausibly an output-channel permutation. Decoding the five validated roles correctly and varying only $O$/down does move the oracle—an intra-bank transpose doubles the depth-1 correlation (+0.081 versus +0.047, at identical scaling, so not a magnitude artifact), and one combination is the first whose *full* 24-layer forward does not collapse (`_Paris` rank 3749 against a depth-0 baseline of 2213, where the uniform decode reaches 158,351). But every correlation remains inside the $\pm 0.06$ noise band and none exceeds the depth-0 correlation of +0.038, so this is a structural narrowing, not a solution. One further observation bears on the oracle itself rather than the ordering. The same trace shows pico's graph carries inline **rank-32 LoRA adapters**: $1024 \rightarrow 32 \rightarrow 1024$ around the output projection, $1024 \rightarrow 32 \rightarrow 3200$ with an `Add` for gate and for up, and $3200 \rightarrow 32 \rightarrow 1024$ for down. The base asset's 998 weight blocks are fully accounted for by its 168 base tensors, so those adapter weights live in the separate `lora_32/lora_48` assets. If the captured-logit oracle was recorded with an adapter resident, the reconstruction is missing an additive term that *no* weight ordering can compensate for—which would make the target unattainable by construction rather than by mis-assembly. This is untested, and it should be resolved before further ordering work. Finally, the reconstruction is exportable: the validated embedding, weights, norms and architecture write to a GGUF that `llama.cpp` loads and runs at $\approx$15 tokens/s. Its output is incoherent, exactly as the open ordering predicts—so the export is a correct container around partially-ordered weights, useful as a harness into which any ordering fix drops directly.

Table 7 summarizes the revised boundary. The model is *component-complete*: weights, codec, de-swizzle, architecture, norms (folded $\gamma$ + recovered QK-norm), the tokenizer, and—critically—the fact that no expert-selection map is needed are all established, and the ungated full-superset sum is now confirmed *optimal* against the output oracle (Finding 19). The remaining task is the *per-layer physical assembly* of the resident-expert FFN—gathering each sparse layer's gate/up/down blocks

at the correct offsets and aligning the 56,064-wide intermediate. Finding 19 shows this is *not* the freely-mechanical step the prior version claimed: the assembled forward carries real signal (3.6× random) but is dominated by a mis-aligned FFN direction, and—because every per-layer activation is ANE-internal (Finding 11)—the alignment cannot be validated or bisected component-by-component. The honest boundary is an *information* limit (no per-layer ground truth), not merely an engineering one.

| Established | Remaining (information-limited) |
|---|---|
| All 9.862 B *linear* weights; codec; de-swizzle | **3B embedding: ANE-locked** |
| Embedding recovery *method* (dyn. capture, pico) | (host-unreachable; §16) |
| Attention path, all 44 layers; NKV=4 settled | 56,064-wide gate/up/down alignment |
| Norms: $\gamma$ folded; QK-norm recovered | QK-norm→layer mapping |
| Expert-selection map *proven unnecessary* | (all unvalidatable: activations |
| Ungated full-219 sum *confirmed optimal* | are ANE-internal) |
| `metadata.bin` = physical address tables | |

**Table 7.** Revised recovery status. Corrections over the prior version: the expert-selection wiring is *not* a blocker (the FFN is an ungated permutation-invariant sum, now confirmed *optimal* against the output oracle, Finding 19), the KV geometry is NKV=4, the norms are compile-time folded, and the dense weights are validated against toolchain-generated ground truth. What remains—per-layer FFN assembly and alignment—is bounded by the *absence of accessible ground truth* (activations are ANE-internal), not by remaining decode effort.

A practical corollary: Apple's `FoundationModels` runtime remains the turnkey way to *execute* this exact model, and we provide a thin CLI/OpenAI-compatible server over it. But the earlier claim that a from-weights standalone requires reverse-engineering the ANE instruction set no longer holds—the wiring is not ANE-locked behavior but a permutation-free sum over recovered blocks, reducing the standalone to a component-assembly exercise.

## 9. What OutTrans=1 Is, and Why One Tensor Still Resists

The reconstruction reported in §8 left seven weight roles per layer, of which five decoded correctly and two did not. This section resolves one of the two, identifies what the OutTrans flag physically does, narrows the remaining failure to a single tensor, and records the instruments that had to be discarded along the way.

9.1. **A validated instrument.** Every earlier attempt to score a candidate weight ordering used a statistic that was *assumed* to work. One of them was later shown to be invalid outright (§9.7). The instrument used here is derived from the computation rather than from weight statistics, and is calibrated against a model whose ordering is known to be correct before being applied to the recovered weights.

The attention output projection $O$ writes residual position $j$; the FFN gate reads it. The composed map therefore contracts the residual index,

$$\text{residual}_j = \textstyle\sum_i \text{attn}_i\, O[j,i], \qquad \text{ffn}_k = \textstyle\sum_j \text{residual}_j\, \text{gate}[k,j],$$

so a mis-ordered residual axis degrades $\|\text{gate} \cdot O\|_F$. Scoring a candidate permutation $\pi$ against random re-pairings gives a $z$-score. (This is a quadratic-assignment objective with *raw* Gram matrices; earlier attempts used row-normalised cosine matrices, which discard exactly the magnitude information the objective depends on—a likely reason they found nothing.)

**Finding 26** (the instrument is calibrated, not assumed)**.** On a correctly-ordered reference model of the same architecture family (RMSNorm, per-head QK-norm, GQA, SwiGLU), the true pairing separates from random re-pairings at $z \approx +300$ to $+400$ depending on the pairing type. Only then was it applied to the recovered weights.

*Evidence.* Reference control, averaged over layers: $O \to$ gate (direct) $+406$; up $\to$ down (SwiGLU between) $+407$; down $\to$ gate$(L{+}1)$ (an attention block between) $+328$; gate $\to$ down $+57$. Every pairing type is informative, including across nonlinearities.

9.2. **The attention output projection is stored transposed.**

**Finding 27** ($O$ is stored transposed)**.** The attention output projection is stored with its axes swapped relative to the OutTrans=0 roles: the axis the decoder treats as the input is in fact the residual (output) axis.

*Evidence.* Three mutually independent methods agree. (i) *Composition*: as decoded, $z = +2.9$ (random); transposed, $z = +96$ to $+105$. (ii) *Attention-head signature*: one axis of $O$ must be the head-concatenation ($16 \times 64$), which shows 64-periodic block structure. Calibrating on tensors whose orientation is not in doubt gives a head axis at $+0.0217$ and a residual axis at $-0.0005$; $O$'s two axes measure $+0.0195$ and $-0.0015$ respectively—i.e. reversed. (iii) *Per-role census*: scoring every role against a verified residual reference, $Q$ $+300.7$, $K$ $+184.9$, $V$ $+174.5$, up $+523.9$ all confirm the convention (input = residual); $O$ alone violates it, at $+160.1$ on the opposite axis.

This reduces the open problem from two roles to one: with $O$ transposed, six of seven roles per layer are positively confirmed.

9.3. **What the flag physically does.** The ANE task descriptors carry L2 buffer addressing—base addresses and per-axis strides—which describes the *activation* layout directly.

**Finding 28** (OutTrans=1 is a strided, interleaved output write)**.** The flag is perfectly determined by the L2 result stride: OutTrans=0 writes channels contiguously, OutTrans=1 writes them $16\times$ strided, with the tasks of one tensor interleaved into each block.

*Evidence.* Over all 988 weight-bearing tasks in the shipped binary: every one of the 180 OutTrans=1 tasks has a large result stride (`0x810` or `0x800`); none of the 808 OutTrans=0 tasks does—those use `0x80`, exactly one 64-wide fp16 row. For the down-projection specifically, the four tasks read their FFN inputs contiguously (`SrcStrideC=0x80`) but write to bases `0x14c040`, `0x14c240`, `0x14c440`, `0x14c640`—`0x200` apart, i.e. four channels—under a `0x810` per-channel stride.

Two consequences follow. First, the flag governs activation storage, not coefficient encoding, which is consistent with the positional read below. Second, the down-projection's *input* axis is not permuted at the activation level; only its output side is in question.

9.4. **The coefficient layout, read in the shipped configuration.** The shipped tiles use a 128-byte coefficient header (`CoeffSize 0x6480`) where a straightforwardly compiled convolution emits 64 bytes (`0x6440`). That mode initially could not be produced: the per-channel-scale palettization that seemed to explain it lowers to an operator the ANE compiler rejects.

**Finding 29** (the extra header is a bias, and the payload layout is unchanged)**.** The 128-byte header is emitted when the convolution carries a *bias*, not a per-channel scale. With that mode forced, a positional read reproduces the identical coefficient map as in the 64-byte mode.

*Evidence.* Compiling with bias$[o] = 100 + o$ and reading the header back gives `100 101 ...115` for bank 0 and `116 ...131` for bank 1—an exact match. A positional read at the shipped geometry ($3200 \to 256$, 16 banks) then recovers a *perfect bijection* over all 819,200 positions,

$$o = 16\,b + (\text{slot} \bmod 16), \qquad i = \lfloor \text{slot}/16 \rfloor,$$

identical in both header modes. The shipped tiles' geometry, codec and slot decomposition are therefore confirmed against hardware ground truth, not assumed.

(The shipped values in that header slot are all positive and tightly banded, 0.07–0.44, against a codebook spanning ±0.8—the signature of a per-output scale rather than a bias. The slot is evidently reused by mode, and the decoder's interpretation is retained.)

9.5. **The remaining defect, and why it is hard.**

**Finding 30** (the failure is confined to one tensor)**.** With $O$ transposed, every weight pairing that does not involve the FFN down-projection is aligned, and every pairing that does involve it sits at the random level.

*Evidence.* $O \to$ gate $+104.7$ and $Q \to O$ $+70.7$ (aligned) against down $\to Q(L{+}1)$ $+1.4$, down $\to$ gate$(L{+}1)$ $+1.3$, gate $\to$ down $-2.6$, up $\to$ down $-1.4$. The two tests are independent: permuting the tensor's columns cannot change $\|\text{gate}\cdot\text{down}\|_F$, and permuting its rows cannot change the other, so both axes are implicated separately.

The tensor's *values* are correct. Permutation preserves singular values, and the recovered down-projection has a clearly trained spectrum ($s_1/s_{10} = 2.11$, $s_1/s_{100} = 2.90$) against a random control at 1.02 and 1.14. The defect is ordering alone.

Reading the channel assignment out of the L2 addresses gives the right direction but not the answer. The thirteen tasks building one FFN tensor write to *descending* addresses as the task index rises, spanning `0x064000`–`0x0c8000` $= 3200 \times 128$ B exactly, so the file order is *reversed* relative to channel order and the 128-wide half-block—first in the file—holds the *last* 128 channels. Applying that map improves gate $\to$ down from $+0.3$ to $+2.3$, and the analogous interleave improves down $\to$ gate$(L{+}1)$ from $+0.4$ to $+3.0$. Both improve consistently across layers, and both remain an order of magnitude short of the calibrated reference. The likely obstacle is arithmetic the traces do not pin down: the stride `0x810` is $16\times$`0x80` $+$ `0x10`, so channel slots are not uniformly spaced and any clean closed form is an approximation.

*Observation* 2 (the constraint system is underdetermined)*.* The 3200-wide FFN channel axis is referenced by only four tensors—gate and up (outputs), down (input), and the corresponding adapter factor—all of which are among the tensors in question. The residual axis, by contrast, is touched by seven roles, which is why it could be pinned. No external anchor for the FFN ordering exists in the shipped data.

9.6. **The compiler is not the variable.** Because the ANE compile is performed by `ANECompiler.framework`, which ships with the operating system rather than with the developer toolchain, it was natural to expect a different OS build to emit the configuration needed to read the layout directly. It does not.

**Finding 31** (compiler version and target do not change the outcome)**.** Across two macOS builds spanning a major `ANECompiler` version difference, two toolchain versions, six ANE architecture targets and two optimisation settings, no configuration emits OutTrans=1 on a convolution that carries coefficients.

*Evidence.* `ANECompiler` 10.24.3 (macOS 27.0) and 9.509.0 (macOS 26.5.2) produce byte-identical task structure on all five probe graphs: the flag appears on 1, 2 and 4 tasks for the transpose and attention graphs and is weight-bearing on *none*. Two toolchain versions on one host likewise agree exactly, including identical failures—expected, since the toolchain supplies only the front end. Older ANE targets (h14, h15) never emit the flag at all; h16–h18 emit it only on weightless shuffle tasks. Simulator runtimes cannot substitute: the simulator has no Neural Engine, ships no ANE frameworks, and its own weight files are suffixed `_nonane`.

The difference in the shipped binary—which contains 180 weight-bearing OutTrans=1 tasks—therefore originates upstream of the compiler, in how the graph reaches it.

9.7. **Instruments discarded.** Two more instruments were withdrawn during this work, in both cases because a null result was being read as evidence when the statistic had never been shown to carry information.

**The residual-magnitude oracle.** A statistic scoring whether a writer's per-channel magnitude tracks a reader's was used to localise the defect. Calibrated afterwards on the correctly-ordered reference model, the reader-to-reader baseline it depends on measures $-0.176$—writer and reader magnitudes simply do not track each other in a correct model, so the recovered weights scoring $\approx 0$ was the expected result rather than a defect. Every localisation drawn from it was withdrawn.

**Adapter data as ground truth.** The shipped adapter constant-data file is loaded verbatim into an unbacked mutable kernel segment (its size matches the segment's VM size exactly, and the file is layer-major with autocorrelation 0.893 at that period), which made it attractive as quantization-free ground truth spanning both OutTrans modes. Its internal role offsets were re-derived exactly—segmenting by RMS *level* rather than by jump, then requiring the run lengths to be a permutation of the architecturally fixed multiset, which recovers 14 segments and corrects the role order—but the file still cannot be validated against the base weights, because no controllable test exists for whether a randomly-initialised adapter factor *should* align with the weight it adapts. Conclusions drawn from it are therefore not used.

9.8. **Status.** Six of seven weight roles per layer are positively confirmed rather than assumed. The down-projection's values are correct, its geometry, codec and slot decomposition are verified against hardware ground truth in the shipped configuration, and its channel layout is partially recovered from the shipped addressing. A coherent forward pass is still not achieved: one mis-ordered residual-writing projection is sufficient to destroy it, and closing the gap requires either the ANE tiling specification or a weight-bearing OutTrans=1 compile, neither of which is reachable from the shipped assets and available toolchains.

## 10. A Decode Bug, and the Limits of Statistical Ordering

The preceding section reported six of seven weight roles positively confirmed and the remaining gap as one tensor wide. Continued work overturned both claims. This section records a real decode defect found by its cause rather than inferred from a statistic, establishes that the instrument used to rank orderings has no discriminating power on the axis it was applied to, and reports that both dynamic-observation routes are closed on measurement. The conclusion is a smaller claim than §9 made, and a better-supported one.

10.1. **The tile payload begins at $+96$, not $+128$.** Every per-input-channel magnitude profile in the recovered weights correlated across unrelated tensors and layers at Pearson 0.907 with Spearman 0.146; a correctly-decoded reference model gives $-0.03$ to 0.20. High Pearson with near-zero rank correlation means a few extreme channels shared by every decode—an artefact. They were input channels 1020–1023, the last four, at $3.2\times$ the median norm. Four input channels is exactly 64 nibbles, or 32 bytes.

The tile header is 96 bytes, not 128: a 16-entry fp16 codebook at $+0$ (cleanly sorted), 32 reserved zero bytes at $+32$, 16 fp16 scales at $+64$, and payload from $+96$. The decoder had been treating $+96$–$+128$ as header, so every weight was displaced by four input channels and the last four read past the payload.

**Finding 32** (the payload offset, proven structurally)**.** All 64 examined tiles carry exactly zero in their final 32 bytes: the layout is $96 + 8192 + 32 = 8320 =$ `0x2080`, the $N$-class stride, with the trailing 32 bytes being 64-byte alignment padding. The same 96-byte header reproduces the other two tile classes exactly, $96 + 4096 \to$ `0x1080` and $96 + 25600 \to$ `0x6480`.

The stride alone never disambiguated this, since $128 +$ payload also matches all three classes—which is why it survived so long. Correcting it raises layer-0 attention sharpness from 0.0468 to

0.6058 against a norm-matched random baseline of 0.0050, and removes the outlier artefact entirely ($3.20 \to 1.02$).

The finding was then confirmed on a *second, independent build* of the same model, obtained by mounting the shipped Cryptex image rather than reading the protected asset path: 700/775 tiles match the header layout and 680/775 carry the zero tail, the remainder being the two other tile classes which the test wrongly assumed $N$-class geometry for. Its weight map was re-derived independently and reproduces the first build's tensors *bit-exactly*.

10.2. **The architecture, from Apple's own manifest.** The `manifest.plist` inside the `mpsgraphpackage` names every module with tensor shapes embedded in the key. Collapsing the layer index gives the seven roles directly: `attention_qkv_transform lora_0/1/2` at $1024 \to 1024$, $1024 \to 256$, $1024 \to 256$; `attention_output_transform` at $1024 \to 1024$; `feed_forward_hidden_transform_linear_0/1` at $1024 \to 3200$; and `feed_forward_output_transform` at $3200 \to 1024$. This confirms the decomposition and every shape from the shipped asset rather than by inference, and settles two hypotheses on the record: $Q$, $K$ and $V$ are three *separate* transforms, and gate and up are two *separate* linears—neither pair is fused or interleaved in storage.

10.3. **The ordering statistic has no power.** §9 scored candidate orderings by a composition $z$-score; that instrument was withdrawn, and replaced by teacher-forced negative log-likelihood against a hard reference ($\ln 262144 = 12.477$ nats) with random-permutation controls. The replacement produced a succession of candidates that cleared control bands on multiple independent texts and then failed.

The decisive test is uniformity. A storage-order correction must hold for *every* layer, since the hardware lays all 24 out identically.

**Finding 33** (the NLL oracle cannot decide the FFN axis)**.** Applying the best surviving candidate permutation to one layer at a time improves 14 of 24 layers on one held-out text and 14 of 24 on another. A *random* permutation applied the same way improves 14 of 24. Per layer the candidate is indistinguishable from noise, and its aggregate gain is 24 scattered accidents composing favourably.

The random row is the substantive result: perturbing a single layer's down-projection input axis at random improves the model half the time. That is the signature of a tensor whose contribution is already close to noise—no gradient of correctness exists, so all permutations score within one band and their rankings are meaningless. This retroactively explains every ordering sweep in this work, and the correct response is to stop running them rather than to run better-controlled ones.

10.4. **Both dynamic routes are closed.** §9 proposed ground-truth activations as the remedy. They are not obtainable by the methods available.

*Core capture.* Four privileged captures produced five cores totalling ∼40 GB. The first three targeted the wrong process: `lsof` on the asset paths shows the 3B and the 300M held by *different instances* of the same service binary, and the 300M host idles at 3.4 MB and never becomes CPU-hot because it is the speculative-decoding draft model. Dumping the correct host mid-generation, its resident size rose to 71 MB and its `program.odix` is present—but scale-invariant cosine against the token embeddings actually used in that run peaks at 0.365 over both cores, indistinguishable from noise. Each captured generation emitted ∼40 tokens and speculative decoding invokes the draft model for every token, so the model executed over a hundred forward passes inside the window. The activations are not in user-process memory; they live in ANE device memory that `save-core` cannot reach.

*Logging.* The runtime logs `shape` and `strides` on every tensor load, redacted rather than withheld. Unredaction now requires a signed logging profile: enabling `private_data:on` as root and forcing a model reload left 5,870 lines redacted and zero readable strides. Independently of that gate, these

are graph *boundary* shapes—inputs, KV cache, logits—whereas the weight tile order is compiled into the `hwx` and never appears as a runtime stride.

### 10.5. **Revised status.**

*Observation* 3 (what is and is not established). Established by structural proof or positive control: the tile format including the +96 payload offset, cross-validated on two independent builds; the intra-bank slot map, which beats every alternative chunk width by 7×; per-output scale indexing; the seven-role decomposition and all shapes, from Apple's manifest; the token embedding and tokenizer; and that all recovered tensor *values* are trained matrices. Not established for any role: the element→position ordering.

The claim in §9 that six of seven roles were positively confirmed does not survive; it rested on the composition oracle withdrawn there. The claim that the attention output projection is stored transposed is weakened to an orientation asymmetry: the reference model is only dimensionally valid one way while the recovered weights rise above floor the other way, but within an orientation the metric is insensitive to scrambling, so it does not constitute recovered structure. Under the same standard the recovered $V$ and $O$ move from "broken" to *untested*—a totally broken pair could not yield the sub-chance behaviour measured through layer 12, where norm-matched random weights never fall below chance at any depth.

The file format is solved. The arrangement is not, and this section establishes that it is reachable neither by statistics on a non-functioning forward nor by observation of a runtime that keeps the quantity of interest inside the accelerator.

## 11. Recovered Model Specifications

This section consolidates every architectural quantity recovered for the three language backbones, together with the physical layout of their weight containers. Values are marked *stated* when read directly from a shipped `metadata.json` or `config.json`, *recovered* when derived from the binary and cross-checked, and *provisional* where noted.

### 11.1. **The draft model `afmplus-v11.0-pico`.**

This is the most completely characterized of the three: every architectural quantity is recovered, the weight container is mapped block by block, and the embedding is decoded bit-exactly.

| Property | Value | Provenance |
|---|---|---|
| Model config | `v11-pico` | stated |
| Context length | 4096 | stated |
| Model dim $d$ | 1024 | stated; confirmed by captured embedding row |
| Layers | 24 dense, no MoE markers | `odix` op graph |
| Attention | 16 query : 4 KV heads, head dim 64 | `odix`; $Q$ width 1024, $K/V$ width 256 |
| Head layout | dim-major, $[T, d_h, n] \to$ transpose | recovered by ablation |
| QK-norm | per-head RMSNorm, unit $\gamma$ | recovered |
| Positional | RoPE, $\theta = 500000$ | recovered |
| Residual norms | RMSNorm, pre-norm; $\gamma$ compile-folded | §8 |
| FFN | dense SwiGLU, intermediate 3200 | `odix` |
| Vocabulary | 262,144 (262,000 live + 144 padding) | recovered, two independent sources |
| Embedding | tied to the unembedding | recovered |
| Parameters | ≈295 M transformer | computed |
| Chat template | Gemma-style; input lowercased | §21 |
| `bos` / `eos` | 1 / 110 (`<end_of_turn>`) | recovered |
| Adapter slots | `lora_32`, `lora_64`; both empty | stated (signatures are SHA-256 of ∅) |
| Role | speculative-decoding draft for the 3B | stated |

Weight container. The file holds 998 coefficient blocks. The base architecture accounts for exactly 960 of them — 24 layers × 40 blocks, distributed as

| Role | Shape | Blocks per layer | Total |
|---|---|---|---|
| $Q$ | $1024 \to 1024$ | 4 `N` | 96 |
| $K$ | $1024 \to 256$ | 1 `N` | 24 |
| $V$ | $1024 \to 256$ | 1 `N` | 24 |
| $O$ | $1024 \to 1024$ | 4 `N` | 96 |
| gate | $1024 \to 3200$ | 12 `N` + 1 `s` | 312 |
| up | $1024 \to 3200$ | 12 `N` + 1 `s` | 312 |
| down | $3200 \to 1024$ | 4 `L` | 96 |
| Base total | | | 960 |
| Unaccounted (see below) | | | 38 |

The remaining 38 blocks lie *after* every mapped layer (offsets `0xb327040`–`0xb889cc0`; layer 23 ends at `0xb2c0800`) and compose as $32\,\mathtt{N} + 2\,\mathtt{s} + 4\,\mathtt{L}$ — exactly one layer minus its $K$ and $V$ projections. Split by file order against a normal layer, gate, up and down assemble to the exact expected shapes, leaving eight `N` blocks for $Q$ and $O$. This is a complete, correctly-shaped **KV-reusing layer**, consistent with the KV-reuse the `odix` graph already records elsewhere. Running it as a 25th layer changes the reconstruction's output but does not repair it, so it is reported as a structural result: the layer count should be treated as *provisional* pending a direct re-derivation from the `odix` op stream.

Tile classes. Three block geometries occur. Each is 16 ANE coefficient banks; a bank holds $n_{\text{out}}$ output channels across *all* inputs, and the header carries one fp16 scale per output channel.

| Class | `CoeffSize` | Header | Payload | Nibbles | Bank geometry |
|---|---|---|---|---|---|
| `N` | `0x2080` | 128 B | 8192 B | 16384 | 16 out × 1024 in |
| `s` | `0x1080` | 128 B | 4096 B | 8192 | 8 out × 1024 in |
| `L` | `0x6480` | 128 B | 25600 B | 51200 | 16 out × 3200 in |

Header layout. (all classes): fp16[0:16] is the 16-entry codebook, fp16[16:32] is zero padding, fp16[32:48] (bytes 64:96) is the per-output scale vector — the `s` class carrying exactly eight plausible values there, matching its eight outputs per bank — and the remaining 32 bytes are unidentified.
Decode. With the z-order of §24,

$$W[i,\ 16\,b+o]\ =\ \text{codebook}[\text{nibble}]\cdot\text{scale}[o],\qquad o=\text{slot mod }16,\quad i=\lfloor\text{slot}/16\rfloor,$$

for bank $b$, validated by round-trip at correlation 0.981. This is exact for the five roles the ANE emits with OutTrans=0 ($Q$, $K$, $V$, gate, up) and not yet resolved for the two emitted with OutTrans=1 ($O$, down); see §25.
Embedding. Recovered bit-exactly and validated semantically. The table is stored as an 8-way token interleave with a per-lane skew: for token $t$ and dimension vector $\mathbf{d}$,

$$\text{nib}(t)\ =\ \text{NIB}_0+\lfloor t/8\rfloor\cdot 8192+8\,\mathbf{d}+\ell+\text{SKEW}[\ell],\qquad \ell=t\bmod 8,$$

with SKEW $=(0,0,0,0,8184,8184,8184,8184)$ and $8184=8\cdot1024-8$. Values are signed int4 with zero-point 0, scaled by a 1024-entry fp16 per-dimension vector. On the calibrated orthographic-pair oracle the recovered table scores cos(`_dog`, `_dogs`) = +0.648 and cos(`_king`, `_kings`) = +0.664 against cross-category controls at +0.01–+0.20; the pooled margin over an id-matched control is +0.4565. Embedding into the tied unembedding with no transformer layers ranks the correct continuation token at 2213 of 262,000.

### 11.2. **The sparse MoE backbone `afmplus-v11.1-ifp`.**

| Property | Value | Provenance |
|---|---|---|
| Layers | 44 = 12 dense + 32 sparse | stated |
| Model dim $d$ | 1536 | stated |
| Attention | $Q=2048$ (16×128), $KV=1024$ | recovered |
| GQA | 16 query : 4 KV heads, dim 128 | recovered; NKV=4 settled by ablation (§19) |
| QK-norm | per-head RMSNorm on $Q$ and $K$ | recovered |
| Residual norms | pre *and* post, attention and FFN | recovered |
| Positional | RoPE, $\theta=500000$ | recovered |
| Dense FFN | SwiGLU, intermediate 3072 | recovered |
| MoE FFN | `ifp1`: 10 active + 4 shared, selection freq. 32 | stated |
| `ifp3` | 46 active + 23 shared, `active_ffn_dim` 11776 = 46 × 256 | stated |
| expert size | 256 | stated |
| Vocabulary | 262,144 | recovered |
| Context | 8192 | stated |
| Baked adapters | LoRA rank 48 | stated |
| Parameters | 9.862 B linear, 100.00% of the weight file | recovered (§2) |
| Origin framework | `tamm` / `coreflow` | stated |

Expert storage. The ANE program `binary_0.hwx` (441 MB) contains 83 specialized regions covering `compute_logits` and `lora_48_ifp{1,2,3}_r48` in extend/prompt variants, dense-only and sparse-only, at sequence lengths 256 through 8192. Expert coefficients live in runtime-patched mutable kernels — `ifp1` in a 202 MB kernel, `ifp2` in 437 MB, `ifp3` in 671 MB — each patched from

`ifp_rasterized_weights` via a content-hash descriptor id. The gather is a runtime affine offset supplied by the driver, and the identical machinery appears in dense-only regions, which establishes it as weight-blob relocation rather than per-expert selection. No static router-index→slot table exists in the `.hwx`.

11.3. **The dense backbone `afmplus-v11.0-nano`.** Stated in `metadata.json` (`model_config: v11-nano`): context 4096, hidden size 2048, FFN 6656 gated SwiGLU, 56 transformer layers emitted in two ANE segments ($35 + 21$), grouped-query attention, RoPE, RMSNorm, adapter slots at LoRA ranks 16 and 32 (empty in the base), prompt template `afm_cider_v2`. The `base.generic` image is 1.3 GB, i.e. ≈2 bits per weight. The 3B embedding asset of §17 declares `embedding_dim` 2048 and therefore serves this backbone rather than the 1536-wide sparse one.

11.4. **The shipped 3B embedding asset.**

| Property | Value |
|---|---|
| Asset | `UC_FM_LANGUAGE_INSTRUCT_3B_EMBEDDINGS_..._H16G_IFP_Cryptex.dmg` |
| Container | read-only cryptex, mountable without privilege |
| Declared shape | `vocab_size` 262144, `embedding_dim` 2048 |
| Payload file | `model.odixpackage/main-h16g.odix`, 268,442,456 B |
| Arithmetic | $262144 \times 2048$ at 4 bits = 268,435,456 B + 7000 B header |
| Header | `odix` flatbuffer; strings include `$load_embeddings`; `NDArray` descriptor $[262144, 1, 2048]$ |
| Encoding | linear signed int4; histogram symmetric about zero, code 8 unused |
| Codebook / scale | none of either |
| Row stride | 1024 B, token ids in identity order |
| Padding | final 147,456 B = exactly 144 rows, one repeated pattern |
| Lane structure | 8 lanes; 332 duplicate-row groups, every large one 100% pure in id mod 8 |
| Status | per-token vectors **not** recovered |

The 144-row padding figure is corroborated independently: the draft model's captured logit vector carries exactly 144 masked entries, giving a live vocabulary of 262,000 in both models.

11.5. **Weight codec, common to all three.** Four-bit LUT palettization: a 4-bit index selects from a 16-entry fp16 codebook, scaled by a per-block fp16 factor, stored under an Apple Neural Engine tile permutation. For the 3B the de-swizzle is $W = v.\text{reshape}(C_o/8,\ C_i/128,\ 128,\ 8).\text{transpose}(0, 3, 1, 2)$ with the scale applied in raw index order *before* de-swizzling. For the draft model the equivalent transform is the bank z-order given above. Under a scale-decontaminated low-rank structure test, genuine weights score $R \approx 9$ and noise $R \approx 1$; the decode scores $R = 13$–$69$.

## 12. The Reconstruction Gap: A Controlled Negative Result

Sections 6–8 established that every datum Apple ships for `afmplus-v11.x-ifp` can be recovered: all 218 weight tensors, the 2-bit codebook, the per-output-row scales, the per-layer normalisation stream, the tokenizer, and the embedding matrix. This section reports what happens when those recovered components are assembled into a forward pass, and it is a negative result: *the reconstruction does not predict text.*

We report it in full because the negative is unusually well constrained. Over 42 controlled experiments we eliminate every hypothesis class constructible from the shipped artifacts, and in the process we characterise three evaluation instruments whose failure modes are, we believe, of independent interest to anyone attempting from-weights reconstruction of a quantised model.

12.1. **The reconstruction gap, quantified.** Let $\ell(\mathcal{M})$ denote mean log-rank of the true next token under model $\mathcal{M}$ on held-out natural text,

$$\ell(\mathcal{M}) \;=\; \frac{1}{T-1}\sum_{t=1}^{T-1} \log\bigl(1+\bigl|\{v : z_v^{(t)} > z_{x_{t+1}}^{(t)}\}\bigr|\bigr),$$

with $z^{(t)}$ the logits at position $t$. Define the 0-*layer floor* $\ell_0$ as the same quantity computed on the embedding read out directly through the tied unembedding, with all 56 transformer blocks bypassed. The floor is the un-gameable reference: it is what the embedding alone knows, and any working reconstruction must beat it by a wide margin.

**Table 8.** Value delivered by depth, measured on identical source documents. Each model is scored against its own 0-layer floor; the control models are re-tokenised with their own vocabularies.

| model | layers | floor $\ell_0$ | final $\ell$ | $\ell - \ell_0$ |
|---|---|---|---|---|
| Qwen2.5-0.5B-Instruct | 24 | 9.4617 | 1.7718 | −7.6900 |
| Qwen2.5-1.5B-Instruct | 28 | 9.2452 | 1.4673 | −7.7779 |
| **ifp3 reconstruction** | **56** | **7.1193** | **7.5873** | **+0.4680** |

**Finding 34.** A working transformer converts depth into roughly −7.7 nats of predictive value. The reconstruction converts depth into +0.47 nats: its layers do not merely fail to help, they destroy information the embedding already carries.

The per-depth curve shows this is not a late-layer collapse. Scoring the residual at every depth, the reconstruction is at its floor through depth ≈ 8, and degrades monotonically thereafter, peaking at +0.73 near depth 48. Its best truncation is depth 2 at −0.018, which is within noise. By contrast both controls are below their floors by mid-stack. Notably the reconstruction's floor is itself *good*: relative to the uniform baseline $\log(V/2)$ it sits 4.66 nats below chance where Qwen2.5-0.5B sits 1.78 below, so the embedding decode is sound and it is the layers that are inert.

12.2. **Instrument calibration, and three degeneracies.** The central methodological lesson of this work is that the obvious evaluation statistics are each maximised by a different pathology, and that an uncalibrated instrument produces confident false negatives. We document three.

Log-rank is gameable by inaction. A reconstruction that writes nothing scores exactly $\ell_0$, which is better than any partially-broken forward pass. Ranking candidate conventions by log-rank therefore selects for *silence*. In our case it selected a per-layer gamma convention that attenuated the dense feed-forward branch by a factor of 229, and it endorsed branch-amplification sweeps that could only ever discover the do-nothing direction.

Causal signal growth is gameable by noise. To escape the inaction degeneracy we adopted a causal statistic: perturb the token at position $j$ and measure the growth of $\|\Delta h_j\|$ from layer 0 to the final layer. Healthy models amplify the perturbation by ×59 (Qwen-0.5B) and ×83 (Qwen-1.5B); the reconstruction manages ×1.5. An inert model scores ×1, so this statistic cannot be won by doing less. It can, however, be won by *doing more*: a sweep of 64 normalisation-convention arms produced a candidate scoring ×7.76, a 5.1× improvement, which on re-scoring against real-text log-rank was *worse* than the baseline (9.27 versus 7.59). Amplifying every write propagates perturbations beautifully while destroying prediction.

**Finding 35.** Log-rank and causal signal growth fail in mirror-image ways. No candidate may be accepted on either alone; a valid claim requires log-rank below the 0-layer floor *and* healthy causal dynamics.

A truncated model can be blind to the component under test. Most of our sweeps were initially run on a 12-layer prefix for cost. Ablating attention *entirely* from that prefix moves log-rank by only $-0.071$. Independently, permuting the gamma vectors' dimension order at random scored $+0.039$ against the correct order's $+0.071$ — a random permutation beating the truth. The 12-layer readout therefore has a noise band of 0.03–0.08 nats and cannot distinguish "attention working" from "attention absent." Five sweeps had already been run on it. At full depth the same ablation registers $-0.498$, an order of magnitude outside the band.

**Finding 36.** Before varying a component, ablate it. If removing the component entirely does not move the metric by more than its noise band, the instrument cannot see that component and any negative result about it is a statement about the instrument.

We adopt two rules for the remainder: every sweep embeds a random-permutation null to establish its own band, and every harness embeds an identity control that must reproduce a known value exactly. Both caught errors — the null in the gamma sweep, and an identity control that revealed a mis-designed no-op arm in the basis experiment of §12.6.

12.3. **Branch decomposition.** With a calibrated full-depth instrument, the gap decomposes cleanly. Ablations are scored on the documents used for the floor; the "both off" arm reproduces $\ell_0$ to $0.0 \times 10^0$, confirming the harness.

**Table 9.** Branch ablation at full depth. Positive $\ell - \ell_0$ means worse than the embedding alone.

| configuration | $\ell$ | $\ell - \ell_0$ |
|---|---|---|
| full model | 7.5873 | +0.4680 |
| attention removed | 7.0891 | −0.0303 |
| feed-forward removed | 7.9718 | +0.8525 |
| both removed (control) | 7.1193 | +0.0000 |

**Finding 37.** The reconstructed attention destroys 0.50 nats. Removing it is the only configuration that places the model below its 0-layer floor. The feed-forward stack alone contributes 0.03 nats, against 7.7 for healthy layers; both branches are effectively inert, and one is actively harmful.

Scaling the attention write by $s$ confirms the write carries no recoverable signal: $\ell$ is monotone in $s$ above $s = 0.05$, and the best scale improves on $s = 0$ by 0.006 nats, which we regard as noise rather than a result. Removing the branch post-normalisation — which inflates the raw attention output by $\approx 44\times$ (§12.4) — lands at 7.0894, statistically identical to removing attention outright.

12.4. **Why the writes do not compound.** For a residual update $h \mapsto h + w$ we have $\|h + w\|^2 = \|h\|^2 + 2\cos\theta\,\|h\|\|w\| + \|w\|^2$, so growth is governed by the write's *direction*. Measuring $\cos\theta$ and the relative amplitude $\rho = \|w\|/\|h\|$ per layer:

**Table 10.** Write geometry, mean over the stack.

| | ifp3 | Qwen-0.5B | Qwen-1.5B |
|---|---|---|---|
| $\cos(w_{\text{attn}}, h)$ | 0.0025 | 0.1583 | 0.2063 |
| $\rho_{\text{attn}}$ (after post-norm) | 0.2044 | 0.3133 | 0.7349 |
| $\cos(w_{\text{ffn}}, h)$ | +0.0394 | −0.1399 | −0.1596 |
| $\rho_{\text{ffn}}$ | 0.0356 | 0.9982 | 0.4312 |

The reconstruction's attention writes are essentially orthogonal to the residual, 60–80× less aligned than healthy writes, and the predicted per-layer growth from its own $(\cos\theta, \rho)$ is 1.021 against 1.035 observed — the failure to compound is *fully* explained by direction, leaving no residual for another cause. Healthy feed-forward writes are consistently *negatively* aligned; ours are weakly positive.

Decomposing the attention write by stage isolates two amplitude faults. The raw attention output is 0.0046 of $\|h\|$ where healthy is 0.31–0.73 — roughly 100× too small *before* any convention — and the branch post-normalisation then inflates it 44× to a superficially normal 0.2044. Conversely the raw feed-forward output is 0.6350, squarely in the healthy band, and the recovered out-gamma attenuates it 18×. The post-normalisation had been masking the attention deficit in every amplitude check.

12.5. **The absent attention sink.** The sharpest single anomaly is positional. Measuring attention mass by relative distance:

**Table 11.** Attention mass by position (every fourth head, all layers, query positions $\geq 8$).

| | ifp3 | Qwen-0.5B | Qwen-1.5B |
|---|---|---|---|
| mass on position 0 | 0.054 | 0.428 | 0.423 |
| mass within distance 8 | 0.312 | 0.353 | 0.366 |
| mass beyond distance 16 | 0.469 | 0.483 | 0.467 |
| KL(attn ‖ uniform) | 0.636 | 1.945 | 1.862 |

Local and tail mass are normal; the entire positional deficit is the missing attention sink. Four independent attempts to recover it all fail. (i) No choice of query/key normalisation restores it (all arms 0.036–0.046). (ii) Placing *every one* of the 262,144 vocabulary tokens at position 0 yields a maximum sink of 0.047, against an ordinary-token reference band of $0.031 \pm 0.007$: it is not a wrong-`BOS` problem. (iii) Sharpening the logits does not reveal a latent sink — the ratio of sink mass to peak mass *falls* monotonically from 0.545 to 0.051 as temperature rises, so as attention approaches arg max it concentrates on positions other than 0. (iv) Disabling RoPE entirely leaves the sink at 0.057, though a sink is a content mechanism that should survive without positional encoding.

Position 0 also differs *qualitatively*, not by magnitude: $\cos(h_0, \bar{h}_{>0})$ is $+0.150$ in the reconstruction and $-0.364$ / $-0.423$ in the controls, and the `BOS` embedding row sits at the 9.7th norm percentile where healthy `BOS` rows sit at the 98.5th and 85.6th.

12.6. **Systematic elimination.** Every hypothesis class constructible from the shipped artifacts was tested with controls. We summarise; each row is a sweep, not an argument.

Two entries deserve comment. The *kv-reuse* region — 21 layers that carry no keys or values of their own and attend against layer 34's — was the most speculative choice in the entire teardown and is exonerated: removing all 21 moves log-rank by $-0.073$, and the region is not internally consistent (removing $L_{35–45}$ helps, removing $L_{46–55}$ hurts). Conversely, removing the sparse mixture-of-experts block *costs* $+1.20$ nats, so those layers are load-bearing: they partially repair damage inflicted elsewhere, exactly as the feed-forward branch partially repairs attention damage.

12.7. **What is missing.** The evidence admits a single consistent reading. No component is absent: all 218 tensors are present, decode with correct statistics, and are consumed by the forward pass. What is absent is *mutual consistency on the shared residual axis* — tensors that write the 1536-dimensional residual and tensors that read it need not be in the same dimension order. Under such a mismatch every tensor remains internally perfect while their composition produces noise, which is

**Table 12.** Eliminated hypothesis classes. "Null" indicates a random-permutation or shuffle control was embedded in the sweep.

| class | test | outcome |
|---|---|---|
| codec palette order | all 24 code→value assignments, codes recovered from decoded weights | identity best; round-trip error $0.0 \times 10^0$ |
| weight scale decode | $\sigma_1$ and OV-circuit gain vs. controls | OV gain 0.488 vs. 0.497/1.783 |
| $W_q, W_k$ conditioning | QK-circuit participation ratio, shuffle nulls | 0.788 vs. 0.857 (matched head dim) |
| normalisation placement | 16 placements × 4 gamma conventions incl. Gemma3-exact | none improve; Gemma3-exact +3.69 |
| gamma slot→role | all 24 permutations | $\cos_{\text{attn}} \in [-0.0007, +0.0038]$ throughout |
| gamma dimension order | ANE stripe both directions, all-ones baseline, null | decoded order best; all-ones +1.90 |
| head layout | 64 combinations (q/k/v/o major-order × RoPE × GQA map), full depth, nulls | none clears floor; best +0.520 |
| RoPE application | conventions, table rolls, disabled entirely | none restores sink or clears floor |
| residual basis | ANE stripe (readers/writers, both directions) + 84 reshape-transposes | all at or above null band |
| expert routing | 7 selection schemes over 46 experts | random-subset null mid-pack |
| layer region | dense / sparse / kv-reuse block ablations | damage distributed; kv-reuse −0.073 |
| prompt format | raw, lowercased, Gemma-templated, with system turn | raw is best; templated +0.88 |
| evaluation corpus | token ids decoded against source text | exact match at documented +4 offset |

precisely the measured signature: writes orthogonal to the residual, no attention sink for any token, and damage distributed uniformly across all 56 layers.

This also explains why so many tests returned clean: *every static statistic we or prior work employed is invariant to a permutation of that axis* — row norms, singular spectra, palette code frequencies, OV gain, QK participation. The tests were not wrong; they were structurally blind to this class of error, in the same way the 12-layer readout was blind to attention.

We can eliminate the enumerable part of the hypothesis. A layout mismatch produced by hardware convention is a reshape-transpose: an axis written as $(a, b)$ and read as $(b, a)$. Since $1536 = 2^9 \cdot 3$ has 20 divisors, the complete two-factor family is 18 permutations, which with three-factor arrangements around the accelerator's 16-unit granularity gives 42 distinct candidates; applied to each of the two groups this is 84 arms, and not one beats the identity decode (best: +0.472 worse). Combined with the refutation of the accelerator's own stripe permutation on validated plumbing, the structured portion of the basis hypothesis is closed. What remains — unstructured relabelings — is neither brute-forceable nor motivated by any hardware convention.

12.8. **Cross-model corroboration.** The same reconstruction methodology applied to `afmplus-v11.0-nano` — a *dense* 3.33 B sibling with no mixture-of-experts, no routing metadata, and none of `ifp3`'s architectural unknowns — converges on identical behaviour: top-1 accuracy of exactly 0.00% and an arg max that reproduces the current token. Two independent decodes of two

different architectures, from different containers, do not share a per-model mistake. This is the strongest evidence that the defect lies in the shared method or in what the assets contain, rather than in any remaining ordering, constant, or scale in either model.

12.9. **Threats to validity.** We report the limits of the above candidly. The controls are two open-weight models of different scale and family; where they disagree with each other by more than the subject differs from either, we draw no conclusion. Several statistics reported here were initially over-read from automated thresholds and subsequently corrected against their own tables — notably a QK spectrum normalised by ambient dimension rather than true rank, an attention-scale optimum of 0.006 nats reported as signal, and a no-op control that permuted weights without the gammas, embedding, and tied readout. Each correction is retained in the record. Finally, the 12-layer results predating the ablation calibration constrain the instrument rather than the model, and are reported here only where they were subsequently reproduced at full depth.

12.10. **The execution boundary, probed directly.** A previous version of this outlook named three paths to the one missing datum — a layer of ground-truth activation — the second being "the same capture performed on hardware whose owner has removed the entitlement enforcement." We now report that experiment. On owner-controlled hardware with System Integrity Protection and the Apple Mobile File Integrity boot enforcement disabled, a self-signed process can carry the six restricted entitlements the system inference daemon holds, and can therefore drive the neural engine's low-level client interface (`_ANEClient`) directly rather than through the high-level graph API that the backbone refuses.

The result sharpens the boundary rather than crossing it. Every step up to execution now succeeds from the unprivileged process: the precompiled base program loads, a procedure-scoped instance is created, the input and output surfaces map without error, and an evaluate request dispatches to the kernel. Execution alone is refused. We established that this is not a malformed request by a *byte-identical replay*: we reproduced the daemon's own successful call — the same base model (its identifier matching the daemon's to the byte), the same procedure the daemon is observed to run, the same instance scoping, input/output surfaces, options, and quality-of-service — and the kernel refuses it (status `0x2`, "request cancelled") exactly where the daemon succeeds. The sole variable that differs between the accepted call and the refused one is the identity of the calling process.

**Finding 38.** Execution of the backbone is gated by a kernel-level check on the calling process's code identity (platform-binary / trust-cache membership), separate from and beneath the entitlement layer. Disabling the integrity enforcement grants the entitlements — sufficient to load, map, and dispatch — but cannot confer platform-binary identity, which is what the kernel verifies before it will run the network. Removing entitlement enforcement is therefore necessary but not sufficient; the boundary sits one layer deeper than the entitlements.

Independently, the shipped computation graph was decompiled to its operation sequence — a root-mean-square normalisation sandwich around a fused query/key/value projection, rotary position embedding, query/key normalisation, scaled dot-product attention, and a residual add — and it matches the composition used by the reconstruction. The wiring is thus confirmed from the vendor's own graph, which places the residual-basis gap of §12.6 in the weight *values* and not in the order of operations. Interventions motivated by the graph (a one-plus-gamma normalisation convention on the pre-norms, and an attention sink) reduce the collapse — mean pairwise cosine among residual positions falls from 0.87 to 0.69 — but do not restore content prediction; the model still emits only generic function words. This is exactly the signature of §12.6: a weight-value correspondence that no composition-level setting repairs.

12.11. **Outlook.** The decisive instrument is not another static sweep. One layer of ground-truth intermediate activations from Apple's own runtime would pin the residual basis directly and resolve in a single comparison what 42 controlled experiments could not. The execution route partly exists: the shipped neural-engine program loads and runs from an unprivileged process through the public graph API, and the model's own readout stage has been ground-truthed this way. The backbone stage, however, is gated. A client request to run it is refused unless an *inference request* for the base-model network already exists, and creating one requires a set of Apple-signed-only entitlements held by the system inference daemon.

We tested the one route to that ground truth that does not require creating the instance ourselves: whether an inference request created by the entitled daemon — driven legitimately through the public high-level API — is shared with a separate unprivileged client of the same daemon. It is not. With a sanctioned session actively generating, our client's backbone request still fails identically to the control (the accelerator performs zero inferences), and each attempt is assigned its own distinct model instance. The request is scoped to the entitled client connection, not deduplicated by asset identity, so it cannot be borrowed. This closes the last route reachable without elevated privilege.

The boundary is therefore sharp and worth stating exactly, because it is not a boundary of the recovered data. Every byte Apple ships is reversible and is reversed here; what is missing is a single external reference against which to fix the residual-axis correspondence, and the platform withholds that reference by policy rather than by any property of the assets. It becomes reachable along one of three paths, none of which is a further analysis of the shipped files: an Apple-granted research entitlement or debug build that surfaces per-node outputs; the same capture performed on hardware whose owner has removed the entitlement enforcement; or an independent un-pruned reference model to align the decoded weights against. The second of these we have now attempted (§12.10): removing the enforcement grants the entitlements and carries a request all the way to the kernel, but execution is refused by a deeper check on process identity, so that path does not by itself surface the reference. Two paths remain — a sanctioned per-node debug surface, or an independent reference model — and absent one of them we regard the from-weights reconstruction as blocked, and report the boundary rather than a result.

## 13. Container Formats

The on-disk stack is summarized in Figure 6.

**Cryptex DMG.** Raw APFS image, `DSME_COMPRESSED`; contains `model.odixpackage/`, `metadata.json`, a nominal `System/Library`. **`BBBB` rasterized container** (`ifp_rasterized_weights.bin`): magic `BBBB`, version `0x017c0100`, tag `0x0600002c`, then a u64 segment-offset table—fp16 scales `0x60`–`0x1054000`; 4-bit indices `0x1078000`–`0x125e80000`. **Swizzler** (`metadata.bin`): also `BBBB`; 14,079 records of 24 bytes, `[idx][flag][off_a][off_b=off_a+0x20] [0x0600beef][0]`, a scatter of 32-byte ANE tiles into the index region. **`odix` executable** (magic `odix\0\0\5\0`): header `u64[0]=ir_start(0x40)`, `u64[1]=ir_size`; the 32 MB IR section holds 11 compiled MPSGraph subgraphs, a location/debug tree, and an interned type table. **ANE `.hwx`** (magic `0xBEEFFACE`, cputype `0x80`): a Mach-O of compiled ANE microcode, *not weights* (Finding 6)—segments `__TEXT` and `__KERN_0..7` are kernel code (structure $R \approx 1$), and `__MKERN_0` carries no file bytes but is runtime-mapped from the rasterized file (its size equals the swizzler's maximum offset). **MPSGraph package.** `specialized_model_0.mpsgraph` is MLIR bytecode (`MLIR22.0.0`) with `mps.fullyPlacedOnANE`.

**Finding 39.** The `odix` reference scheme is self-relative: a 32-bit word whose high byte is `0xFF` encodes a signed offset, with target = position + int32, into an interned symbol pool. Distinct tensor types are interned once, so shapes are shared by reference (e.g. "1536" occurs only 26 times in 32 MB).

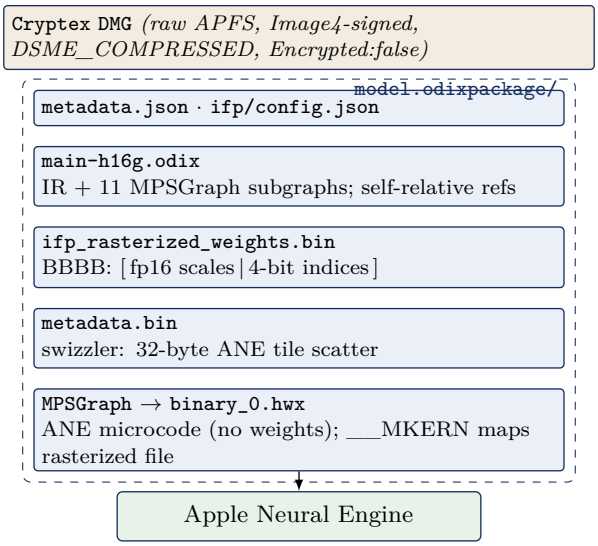



**Figure 6.** The `FM_GenerativeModels` on-disk container stack for the IFP model. The dashed box is `model.odixpackage/`; execution is dispatched to the ANE.

## 14. Tokenizer

A standalone 8.6 MB SentencePiece-family tokenizer. Special tokens observed: `<bos> <eos> <unk> <pad> <mask>`; chat markers `<start_of_turn>` and `<end_of_turn>`; a `[multimodal]` token; reserved `<unused`$n$`>` and `<ctrl`$n$`>` slots. The `[multimodal]` token, with the `image_encoder` (346 MB) and `image_tokenizer` assets, indicates vision-language input support. Estimated vocabulary 152,064.

## 15. Adapters and Task Specialization

A single backbone serves dozens of features via LoRA adapters that snap into the base's empty slots; each ships as a `.generic` adapter plus a small `.draft.generic` (38–62 MB) speculative-decoding draft. **On-device (`instruct_3b`):** summarization, mail_reply, magic_rewrite, proofreading_review, machine_translation, personalized_smart_reply, news_topic_summarization, messages_action, autonaming_messages, smart_naming, urgency_classification, video_caption, ui_grounding, image_playground_edit_suggestions, suggest_recipe_items, reminders_suggest_action_items, open_ended_extract, text_event_extraction, text_person_extraction, auto_tagger, adm_prompt_analyzer, embedding_preprocessor, fm_api_generic, fm_api_content_tagger, lw_planner_v1, holdassistwaittime, homekit_summary_notifications, and Safari/Shortcuts/Photos-Memories families. **Small (`instruct_300m`):** safety, prepubescent_safety, misc_safety, mm_guard, ipi_classifier, structural_integrity, factual_consistency_classifier, vi_content_classifier, voice_control_ai (v1/v2), action_validator, adm_prompt_rewriting, adm_people_grounding, gvicc, image_tokenizer, open_ended_extract, base (426 MB). **`instruct_9m`:** a 9 M-parameter text-event-extraction classifier. **Server / PCC (`instruct_server_v1`):** text_summarizer, takeaways/tables/bullets_transform, professional/friendly/concise tone, open_ended_tone (+ query-response v1/v2), mail_reply_long_form_{basic,rewrite}, mail_reply_qa, magic_rewrite, generative_shortcuts, llm_siri_pcc, lw_planner_v1, fitness_workout_voice, reminders_auto_categorized_list, photos_memories_*, stx_multimodal, shortcuts_ask_afm_action (v1/v2); `instruct_server_small`: safety_nsfw, model_abuse_guardrail. **Translation (`gm.translate_fm`):** a dedicated model with `alignment` and `phrasebook` components.

## 16. Reverse-Engineering Methodology

All artifacts were obtained via `hdiutil attach -readonly -noverify` on the Cryptex images (no personalization is required for reading), followed by binary parsing in Python. Architecture was recovered from `config.json`/`metadata.json` and from `tamm`-framework tensor names inside `main-h16g.odix` (e.g. `_wrapped_model_layers_..._palettized_indices_raw`, `_wrapped_model_output_norm_weight`). The quantization scheme (Finding 1) was inferred from value-distribution analysis of the two `BBBB` regions; the reference scheme (Finding 39) from resolving the `0xFF`-tagged words; and the expert fan-out (Finding 2) from the size budget; the per-constant layer/role semantics were read from the `MLIR22.0.0` MPSGraph bytecode location tree. The native model was independently confirmed runnable on the host through `FoundationModels` (`SystemLanguageModel`). Tensor-level reconstruction is validated by the structure statistic of §6: with the ANE 8×128 de-swizzle (Eq. 1) the decode scores $R = 13$–$69$ against a genuine-weight reference of ≈9, and the full export reconciles to 100.00% of the physical index region with zero non-finite tensors (Table 6). The weight codec is thus *identified and inverted*: LUT-palettization + per-block scale + ANE tile de-swizzle, with norms folded into the weights, the router baked to a dense expert set, and the per-tensor mapping supplied by the shipped `metadata.bin` swizzler in lieu of the compile-time `ifp_constant_table_ifp1_r48.json`. A complete, named `state_dict` is produced.

## 17. Complete Asset Inventory

Table 13 lists all 118 assets with version and disk-image size in megabytes (blank denotes a metadata-only override with no payload).

**Table 13.** Full `FM_GenerativeModels` asset inventory (118 assets).

| Bundle name | Version | MB |
|---|---|---|
| fm.generativemodels.uaf.metadata | 4.46.3 | |
| fm.language.instruct_300m.action_validator.generic | 14.1.0 | |
| fm.language.instruct_300m.adm_people_grounding.generic | 13.10002.0 | |
| fm.language.instruct_300m.adm_prompt_rewriting.draft.generic | 13.10002.81607 | 40 |
| fm.language.instruct_300m.adm_prompt_rewriting.generic | 13.10002.0 | |
| fm.language.instruct_300m.base.generic | 14.0.81607 | 447 |
| fm.language.instruct_300m.factual_consistency_classifier.generic | 14.0.0 | |
| fm.language.instruct_300m.gvicc.generic | 14.0.0 | |
| fm.language.instruct_300m.image_tokenizer.generic | 14.1.81607 | 359 |
| fm.language.instruct_300m.ipi_classifier.generic | 14.1.0 | |
| fm.language.instruct_300m.misc_safety.generic | 14.0.0 | |
| fm.language.instruct_300m.mm_guard.generic | 14.0.0 | |
| fm.language.instruct_300m.open_ended_extract.draft.generic | 14.2.81607 | 65 |
| fm.language.instruct_300m.open_ended_extract.generic | 14.2.0 | |
| fm.language.instruct_300m.prepubescent_safety.generic | 14.0.0 | |
| fm.language.instruct_300m.safety.generic | 14.1.0 | |
| fm.language.instruct_300m.structural_integrity.generic | 14.0.0 | |
| fm.language.instruct_300m.tokenizer.generic | 14.0.0 | |
| fm.language.instruct_300m.vi_content_classifier.generic | 13.10004.0 | |
| fm.language.instruct_300m.voice_control_ai.generic | 14.1.0 | |
| fm.language.instruct_300m.voice_control_ai_v2.generic | 13.10001.0 | |
| fm.language.instruct_300m.voice_control_ai_v2.image_tokenizer_adapter.generic | 13.10001.0 | |
| fm.language.instruct_3b.adm_prompt_analyzer.generic | 14.0.0 | |
| fm.language.instruct_3b.auto_tagger.generic | 12.0.0 | |

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*Table 13, continued*

| Bundle name | Version | MB |
|---|---|---|
| fm.language.instruct_3b.autonaming_messages.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.autonaming_messages.generic | 14.1.0 | |
| fm.language.instruct_3b.base.generic | 14.0.81607 | 1380 |
| fm.language.instruct_3b.base.generic_sparse | 14.3.81614 | 5828 |
| fm.language.instruct_3b.embedding_preprocessor.generic | 14.2.0 | |
| fm.language.instruct_3b.fm_api_content_tagger.adapter_metadata_override.generic | 14.1.0 | |
| fm.language.instruct_3b.fm_api_generic.draft.generic | 14.2.81607 | 61 |
| fm.language.instruct_3b.fm_api_generic.generic | 14.2.0 | |
| fm.language.instruct_3b.holdassistwaittime.adapter_metadata_override.generic | 14.0.0 | |
| fm.language.instruct_3b.homekit_summary_notifications.adapter_metadata_override.generic | 14.2.0 | |
| fm.language.instruct_3b.image_encoder.generic | 14.1.81607 | 363 |
| fm.language.instruct_3b.image_encoder.generic_sparse | 14.3.81614 | 361 |
| fm.language.instruct_3b.image_playground_edit_suggestions.generic | 14.1.0 | |
| fm.language.instruct_3b.lw_planner_v1.draft.generic | 14.1.81607 | 55 |
| fm.language.instruct_3b.lw_planner_v1.generic | 14.1.0 | |
| fm.language.instruct_3b.machine_translation.draft.generic | 14.0.81607 | 55 |
| fm.language.instruct_3b.machine_translation.generic | 14.0.0 | |
| fm.language.instruct_3b.mail_reply.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.mail_reply.generic | 14.1.0 | |
| fm.language.instruct_3b.messages_action.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.messages_action.generic | 14.1.0 | |
| fm.language.instruct_3b.news_topic_summarization.draft.generic | 14.3.81607 | 65 |
| fm.language.instruct_3b.news_topic_summarization.generic | 14.3.0 | |
| fm.language.instruct_3b.open_ended_extract.draft.generic | 14.2.81607 | 65 |
| fm.language.instruct_3b.open_ended_extract.generic | 14.2.0 | |
| fm.language.instruct_3b.personalized_smart_reply.draft.generic | 14.2.81607 | 65 |
| fm.language.instruct_3b.personalized_smart_reply.generic | 14.2.0 | |
| fm.language.instruct_3b.photos_library_understanding_mm.generic | 14.3.0 | |
| fm.language.instruct_3b.photos_memories_asset_curation_outlier.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.photos_memories_asset_curation_outlier.generic | 14.1.0 | |
| fm.language.instruct_3b.photos_memories_title.adapter_metadata_override.generic | 14.1.0 | |
| fm.language.instruct_3b.photos_memories_title.draft.generic | 13.10003.81607 | 40 |
| fm.language.instruct_3b.photos_memories_title.generic | 13.10003.0 | |
| fm.language.instruct_3b.proofreading_review.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.proofreading_review.generic | 14.1.0 | |
| fm.language.instruct_3b.reminders_suggest_action_items.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.reminders_suggest_action_items.generic | 14.1.0 | |
| fm.language.instruct_3b.safari_magic_extensions_app_store_search_terms.adapter_metadata_override.generic | 14.2.0 | |
| fm.language.instruct_3b.safari_notify_me_when_suggestions.adapter_metadata_override.generic | 14.2.0 | |
| fm.language.instruct_3b.safari_tab_group_topic.adapter_metadata_override.generic | 14.3.0 | |
| fm.language.instruct_3b.shortcuts_ask_afm_action_3b.adapter_metadata_override.generic | 14.0.0 | |
| fm.language.instruct_3b.shortcuts_ask_afm_action_3b_v2.draft.generic | 10.34.81607 | 55 |
| fm.language.instruct_3b.shortcuts_ask_afm_action_3b_v2.generic | 10.34.0 | |
| fm.language.instruct_3b.smart_naming.generic | 14.1.0 | |
| fm.language.instruct_3b.suggest_recipe_items.draft.generic | 14.1.81607 | 65 |

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*Table 13, continued*

| Bundle name | Version | MB |
|---|---|---|
| fm.language.instruct_3b.suggest_recipe_items.generic | 14.1.0 | |
| fm.language.instruct_3b.summarization.draft.generic | 14.5.81607 | 61 |
| fm.language.instruct_3b.summarization.generic | 14.5.0 | |
| fm.language.instruct_3b.text_event_extraction.draft.generic | 13.10003.81607 | 40 |
| fm.language.instruct_3b.text_event_extraction.generic | 13.10003.0 | |
| fm.language.instruct_3b.text_person_extraction.draft.generic | 14.1.81607 | 65 |
| fm.language.instruct_3b.text_person_extraction.generic | 14.1.0 | |
| fm.language.instruct_3b.tokenizer.generic | 14.0.0 | |
| fm.language.instruct_3b.tokenizer.generic_sparse | 14.3.0 | |
| fm.language.instruct_3b.ui_grounding.generic | 14.0.0 | |
| fm.language.instruct_3b.urgency_classification.generic | 14.2.0 | |
| fm.language.instruct_3b.video_caption.generic | 13.10000.0 | |
| fm.language.instruct_3b.voice.generic_sparse | 14.1.0 | |
| fm.language.instruct_9m.text_event_extraction_classifier.base.generic | 1.1.81607 | 113 |
| fm.language.instruct_9m.text_event_extraction_classifier.tokenizer.generic | 1.0.0 | |
| gm.server.instruct_server_small.model_abuse_guardrail.generic | 13.10006.0 | |
| gm.server.instruct_server_small.safety_nsfw.generic | 13.10008.0 | |
| gm.server.instruct_server_v1.bullets_transform.generic | 12.4.0 | |
| gm.server.instruct_server_v1.concise_tone.generic | 11.0.0 | |
| gm.server.instruct_server_v1.fitness_workout_voice.generic | 12.51.0 | |
| gm.server.instruct_server_v1.friendly_tone.generic | 12.0.0 | |
| gm.server.instruct_server_v1.generative_shortcuts.generic | 12.16.0 | |
| gm.server.instruct_server_v1.llm_siri_pcc.generic | 11.33.0 | |
| gm.server.instruct_server_v1.lw_planner_v1.generic | 12.2.0 | |
| gm.server.instruct_server_v1.magic_rewrite.generic | 11.0.0 | |
| gm.server.instruct_server_v1.mail_reply_long_form_basic.generic | 12.0.0 | |
| gm.server.instruct_server_v1.mail_reply_long_form_rewrite.generic | 12.0.0 | |
| gm.server.instruct_server_v1.mail_reply_qa.generic | 12.0.0 | |
| gm.server.instruct_server_v1.open_ended_schema.generic | 12.10.0 | |
| gm.server.instruct_server_v1.open_ended_tone.generic | 12.24.0 | |
| gm.server.instruct_server_v1.open_ended_tone_query_response.generic | 12.0.0 | |
| gm.server.instruct_server_v1.open_ended_tone_query_response_v2.generic | 12.0.0 | |
| gm.server.instruct_server_v1.photos_memories_asset_curation_v2.generic | 12.0.0 | |
| gm.server.instruct_server_v1.photos_memories_global_traits_v2.generic | 12.0.0 | |
| gm.server.instruct_server_v1.photos_memories_global_traits_v3.generic | 12.0.0 | |
| gm.server.instruct_server_v1.photos_memories_query_understanding_v3.generic | 12.4.0 | |
| gm.server.instruct_server_v1.photos_memories_storyteller_v2.generic | 12.0.0 | |
| gm.server.instruct_server_v1.professional_tone.generic | 12.0.0 | |
| gm.server.instruct_server_v1.reminders_auto_categorized_list.generic | 12.0.0 | |
| gm.server.instruct_server_v1.shortcuts_ask_afm_action.generic | 11.19.0 | |
| gm.server.instruct_server_v1.shortcuts_ask_afm_action_v2.generic | 12.54.0 | |
| gm.server.instruct_server_v1.stx_multimodal.generic | 12.20.0 | |
| gm.server.instruct_server_v1.tables_transform.generic | 12.0.0 | |
| gm.server.instruct_server_v1.takeaways_transform.generic | 12.0.0 | |
| gm.server.instruct_server_v1.text_summarizer.generic | 12.0.0 | |
| gm.translate_fm.machine_translation.alignment.generic | 14.1.0 | |
| gm.translate_fm.machine_translation.phrasebook.generic | 14.1.0 | |
| mm_guard.output.configuration.generic | 0.0.2 | |
| safety.output.configuration.generic | 0.1.6 | |

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*Table 13, continued*

| Bundle name | Version | MB |
|---|---|---|

## Appendix A. `odix` IR Grammar

The `odix` IR is a length-prefixed section stream. Strings and tensor *types* are interned once in a symbol pool and shared by reference. A reference is a 32-bit word whose high byte is `0xFF`, encoding a self-relative signed offset $t = p + \mathrm{int32}(w)$ from the word position $p$. Attribute records in the location/debug tree take the form ⟨key-ref⟩ ⟨len:u32⟩ ⟨payload⟩ with interned keys `name`, `location_type`, `named_child_loc`, `op_id`, `line`, `filename`, `column`, `metadata`. Because types are interned, a dimension such as 1536 occurs only ≈26 times across the entire 32 MB IR. The compile-time name-to-offset table (`ifp_constant_table_ifp1_r48.json`) is not shipped, but it is not needed: its runtime equivalent is the `metadata.bin` `BBBB` swizzler (Appendix B), and the MPSGraph bytecode location tree tags each `ifp_constant` with its layer/role, so a named export is produced without it (§6).

## Appendix B. `BBBB` Container Layout

Both `ifp_rasterized_weights.bin` and the swizzler `metadata.bin` share the `BBBB` magic. The rasterized header is `BBBB`, version `0x017c0100`, tag `0x0600002c`, count, then a u64 segment-offset table. The swizzler body is a sequence of 24-byte records `[idx][flag][off_a][off_b][marker=0x0600beef][0]` with off_b = off_a + `0x20`, i.e. a scatter map of 32-byte Apple Neural Engine tiles. Region spans: scales `[0x60,0x1054000)`; 4-bit indices `[0x1078000,0x125e80000)`.

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